JGS
Research article
Journal of the Geological Society
Published Online First
https://doi.org/10.1144/jgs2018-076

Late Cretaceous subduction-related magmatism on the southern edge of Sabzevar basin, NE Iran

Zakie Kazemi[1]*, Habibollah Ghasemi[1], Romain Tilhac[2], William Griffin[2], Hadi Shafaii Moghadam[3], Suzanne O'Reilly[2] & Fardin Mousivand[1]

[1] Department of Petrology and Economic Geology, Faculty of Earth Sciences, Shahrood University of Technology, Shahrood, Iran
[2] ARC Centre of Excellence for Core to Crust Fluid Systems and GEMOC ARC National Key Centre, Earth and Planetary Science, Faculty of Science and Engineering, Macquarie University, Sydney, NSW 2019, Australia
[3] School of Earth Sciences, Damghan University, Damghan, Iran
W.G., 0000-0002-0980-2566; H.S.M., 0000-0002-3674-9764
* Correspondence: Sahar_Kazemi6464@yahoo.com

Abstract: Despite detailed geological studies on the Late Cretaceous Sabzevar ophiolites, the non-ophiolitic, subduction-related igneous rocks of NE Iran are still enigmatic. This paper focuses on the Late Cretaceous, non-ophiolitic Sabzevar magmatic rocks, and presents both precise U–Pb ages and a tectonomagmatic scenario for the genesis of these igneous rocks. A thick sequence of Late Cretaceous acidic volcanic rocks associated with sandstones, shales and pelagic limestones dominates the northern parts of the Lut block. This sequence is intruded by a series of shallow mafic to felsic intrusions. New zircon U–Pb results on the intrusive and extrusive rocks show ages of 75–101 Ma. These ages show a pulsed magmatism in NE Iran, which lasted for *c.* 25 myr. The whole-rock geochemistry of the igneous rocks shows suprasubduction-zone-related geochemical signatures, represented by negative anomalies in high field strength elements and enrichment in large ion lithophile elements. The initial $^{87}Sr/^{86}Sr$ ratios and εNd(t) values of the extrusive rocks range from 0.70411 to 0.70628 and from +5.9 to +7.4 respectively, and for intrusive rocks are in the range of 0.70423–0.70579 and +5.8 to +7.2. High εNd(t) values for these rocks confirm that their melts were derived from a depleted-mantle source. Both geochemical and isotopic data indicate that the genesis of these rocks was related to the partial melting of mid-ocean ridge basalt-type slab or depleted-mantle wedge sources during the northward movement of the Sabzevar oceanic slab beneath the southern edge of Eurasia (Turan block) in Late Cretaceous time.

Received 11 April 2018; **revised** 3 December 2018; **accepted** 8 January 2019

Iran lies in the middle of the Alpine–Himalayan belt, the world's most active intra-continental orogen, at the convergence point of the major Arabian and Eurasian plates. As part of this belt, Iran has received special attention and has served as a test-bed for different ideas as geological paradigms have changed. As a result of the development of the Neotethys Ocean, which included the opening, subduction and closure of the ocean as well as post-collision processes, a series of magmatic events has occurred in Iran, one of the most important of which occurred during the Late Cretaceous.

Cenozoic igneous rocks of Iran encompass a region that is up to 1000 km broad from east to west and north to south (Fig. 1), with a magmatic peak in Eocene time, in the Urumieh–Dokhtar Magmatic Belt, a 50–80 km wide magmatic belt that trends NW–SE for 1000 km from NW to SE Iran. The Urumieh–Dokhtar Magmatic Belt evolved for *c.* 100 myr, beginning with the modern phase of subduction starting in Late Cretaceous time and continuing through Cenozoic time.

In general, the Late Cretaceous magmatism was less extensive than the Eocene high magmatic fluxes and crops out in the Urumieh–Dokhtar Magmatic Belt (e.g. Hosseini *et al.* 2017), as well as in NE Iran (e.g. Alaminia *et al.* 2013) (Fig. 1). Alaminia *et al.* (2013) considered the northward subduction of the Sabzevar oceanic lithosphere beneath Eurasia to be responsible for the generation of the Late Cretaceous magmatism in NE Iran. Late Cretaceous magmatism is also common in NW Iran (in Sanandaj), reflecting the NE-ward subduction of the Neotethyan Ocean beneath Central Iran during the Late Mesozoic (Azizi & Moine Vaziri 2009; Agard *et al.* 2011).

Eocene magmatism is widespread in Iran, including Central Iran (Urumieh–Dokhtar Magmatic Belt), the southern margin of central–eastern Alborz, western Alborz–Azerbaijan and within the Lut block (Chiu *et al.* 2013; Pang *et al.* 2013; Ghasemi & Rezaei 2015; Jamshidi *et al.* 2015; Moghadam *et al.* 2016; Yousefi *et al.* 2017) (Fig. 1). Many researchers consider that the Urumieh–Dokhtar magmatic activity was related to the Neotethys subducted lithosphere, because of the spatial relationships between the Neotethys oceanic basin and the continental margin of South Eurasia (Berberian & King 1981). In terms of spatial and structural trends, the Urumieh–Dokhtar magmatic belt runs parallel to the Zagros orogen and the Sanandaj–Sirjan metamorphic–magmatic zone, and the location of the Neotethys subduction zone makes it logical that these belts formed as a magmatic arc above the Neotethys subducted lithosphere. Some researchers believe that the Eocene magmatism in the Sabzevar zone was related to the continuing subduction of the Sabzevar oceanic lithosphere (Alaminia *et al.* 2013; Ghasemi & Rezaei 2015; Moghadam *et al.* 2016; Yousefi *et al.* 2017). The occurrence of magmatic rocks in eastern Iran is enigmatic. Many researchers believe the subduction of the Sistan ocean to the east beneath the Afghan block during Late Cretaceous time was responsible for the genesis of Cretaceous rocks in eastern Iran (e.g. Barrier & Vrielynck 2008). In addition, genesis of the post-collisional Late Eocene to Plio-Quaternary magmatism in eastern Iran is suggested to be related to lithosphere delamination (Pang *et al.* 2013).

It is important to understand the tectonomagmatic evolution of the Iran arcs during the Late Cretaceous, which seems to be diverse

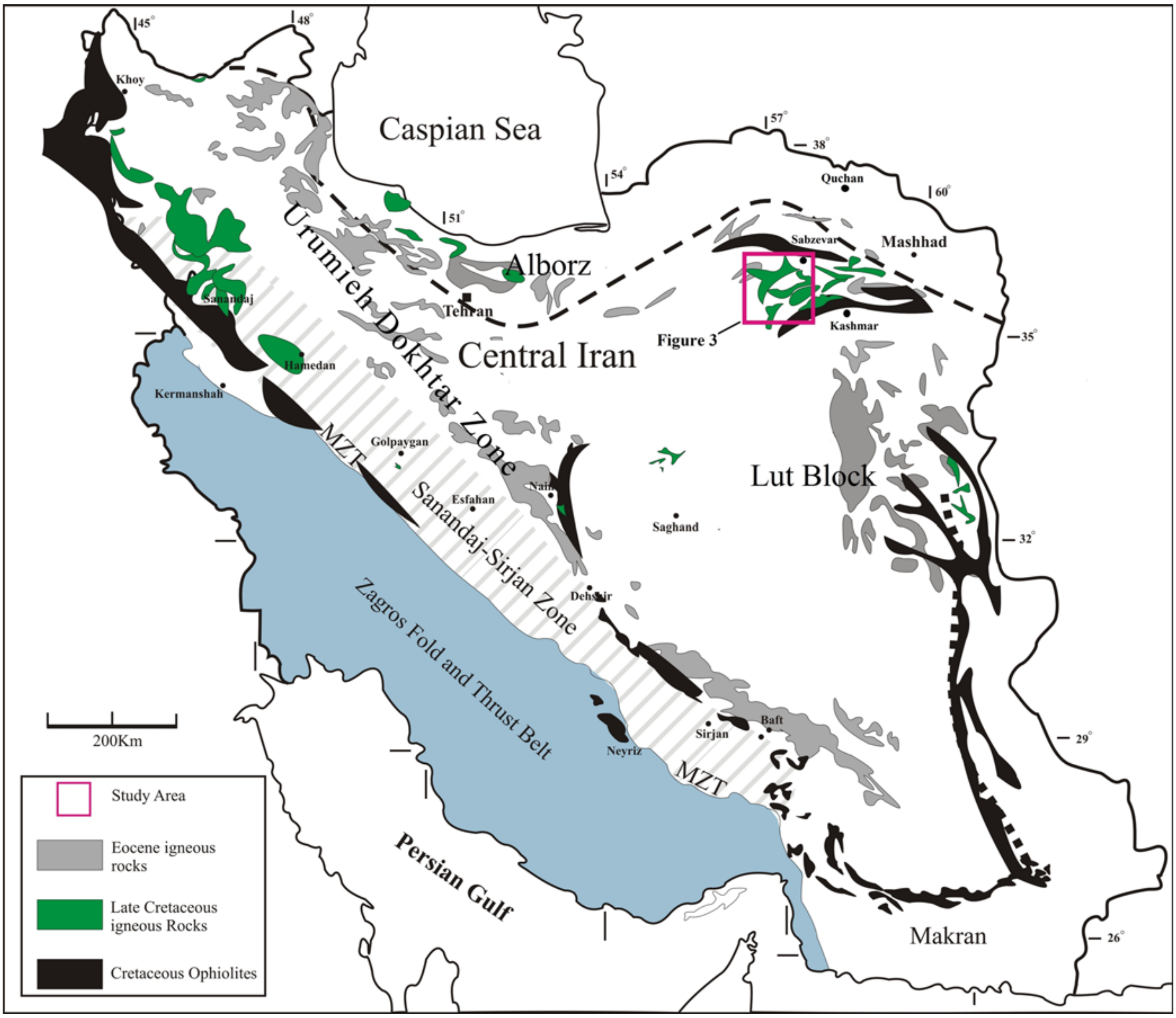

Fig. 1. Map showing the distribution of Late Cretaceous and Eocene igneous rocks. MZT, Main Zagros Thrust.

in different parts of Iran. The Late Cretaceous magmatism in NE Iran is enigmatic and no detailed study has been done yet to understand its genesis mechanism and sources. The general belief is that the subduction of the Sabzevar oceanic crust (as a late Cretaceous back-arc basin) beneath the Turan block resulted in magmatism along the Sabzevar zone during the Late Cretaceous–Eocene (Rossetti *et al.* 2010; Alaminia *et al.* 2013; Jamshidi *et al.* 2015; Maghfouri *et al.* 2016). However, the timing and magmatic products of subduction in the Sabzevar basin are still questionable. We do not know the exact age of subduction initiation in the Sabzevar ocean and how many years it lasted. What were the magmatic products of this subduction and how were they formed? With this in mind, we have collected and precisely dated the igneous rocks that represent the subduction event in the Sabzevar basin. It is also valuable to systematically and accurately study the petrogenesis and tectonomagmatic setting of the Late Cretaceous volcanic rocks and sub-volcanic masses (dykes, sills and stocks) from the Sabzevar zone, NE Iran.

In this contribution, we present new zircon U–Pb age dating, Sr–Nd isotopic data, and mineral and whole-rock geochemistry of the late Cretaceous igneous rocks from the SW Sabzevar area, NE Iran. These new data allow us to determine the magma sources, the path of magmatic evolution and the tectonomagmatic setting of the Sabzevar zone during the Late Cretaceous, and to provide a new model for the tectonic evolution of the region.

Geological setting

The geological structure of Iran can be discussed in terms of seven tectonomagmatic units, from undeformed trench fill in the Persian Gulf (in the SW) (Fig. 1), passing through a consolidated Neoproterozoic crust in Central Iran and to a volcanic belt–geosuture in NE Iran. These units from SW to NE include (1) Zagros fold-and-thrust belt, (2) Outer Belt Zagros ophiolites, (3) Sanandaj–Sirjan Zone, (4) Inner Belt Zagros ophiolites, (5) Urumieh–Dokhtar Magmatic Belt, (6) the uplifted late Neoproterozoic crust of Central Iran and (7) late Cretaceous Sabzevar ophiolites and their accompanying non-ophiolitic late Cretaceous to Cenozoic arc-related rocks.

The northward subduction of the Sabzevar oceanic lithosphere and its relevant Mesozoic magmatism, especially in the Cretaceous, has been widely discussed over the last four decades (Lindenberg *et al.* 1983; Spies *et al.* 1983; Shojaat *et al.* 2003; Rossetti *et al.* 2010; Omrani *et al.* 2013; Moghadam *et al.* 2014; Ghasemi & Rezaei 2015; Jamshidi *et al.* 2015; Maghfouri *et al.* 2016; Moghadam *et al.* 2016). However, most studies have focused on

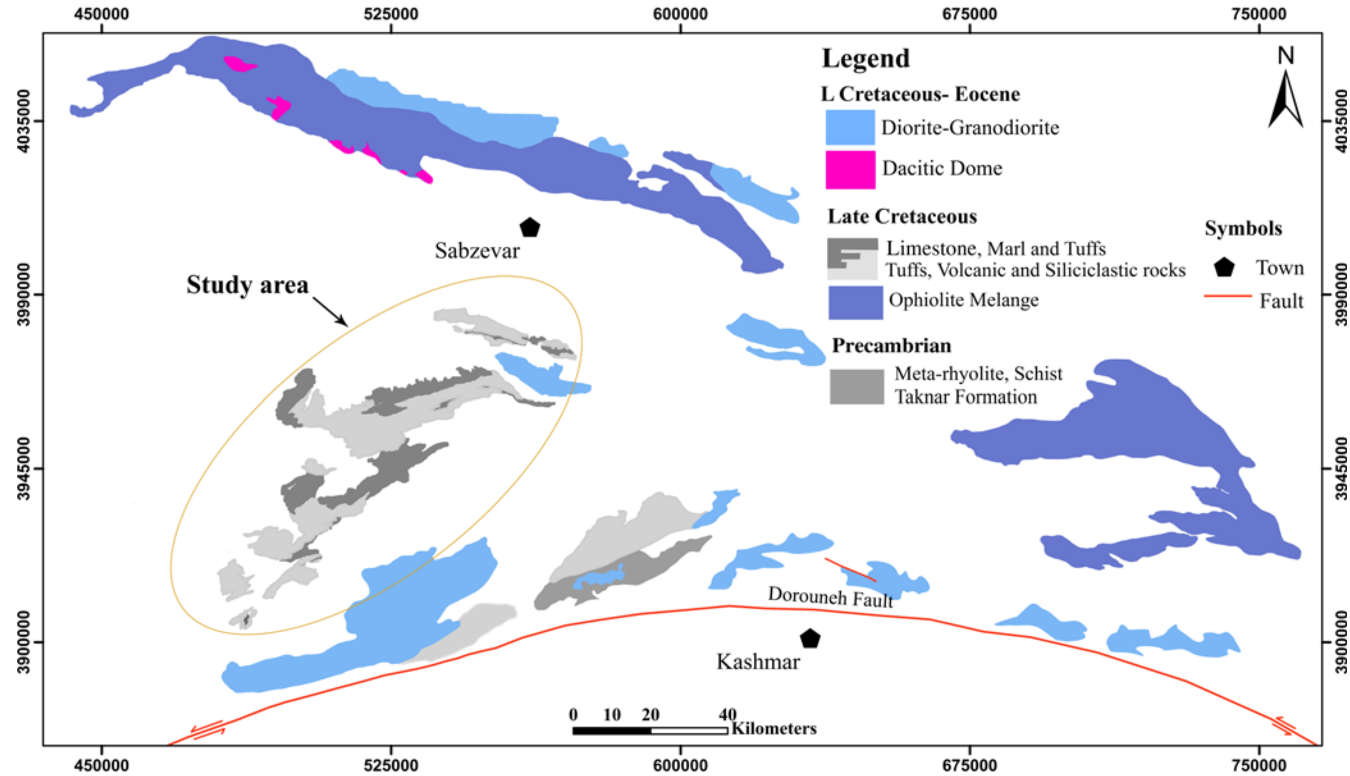

Fig. 2. Geological map of Sabzevar region with emphasis on the distribution of ophiolitic and volcanic rocks.

the petrology, geochronology and geochemistry of the Sabzevar ophiolites and associated Cenozoic volcanic sequence in the northern part of Sabzevar, and there is little information about the Late Cretaceous non-ophiolitic magmatism of the SW Sabzevar area.

The Sabzevar ophiolites (NE Iran), are considered as remnants of Mesozoic back-arc oceanic basins, formed behind the Zagros ophiolites as a result of extreme extension in the Iranian plateau during late Cretaceous time (Fig. 2). The extreme extension was

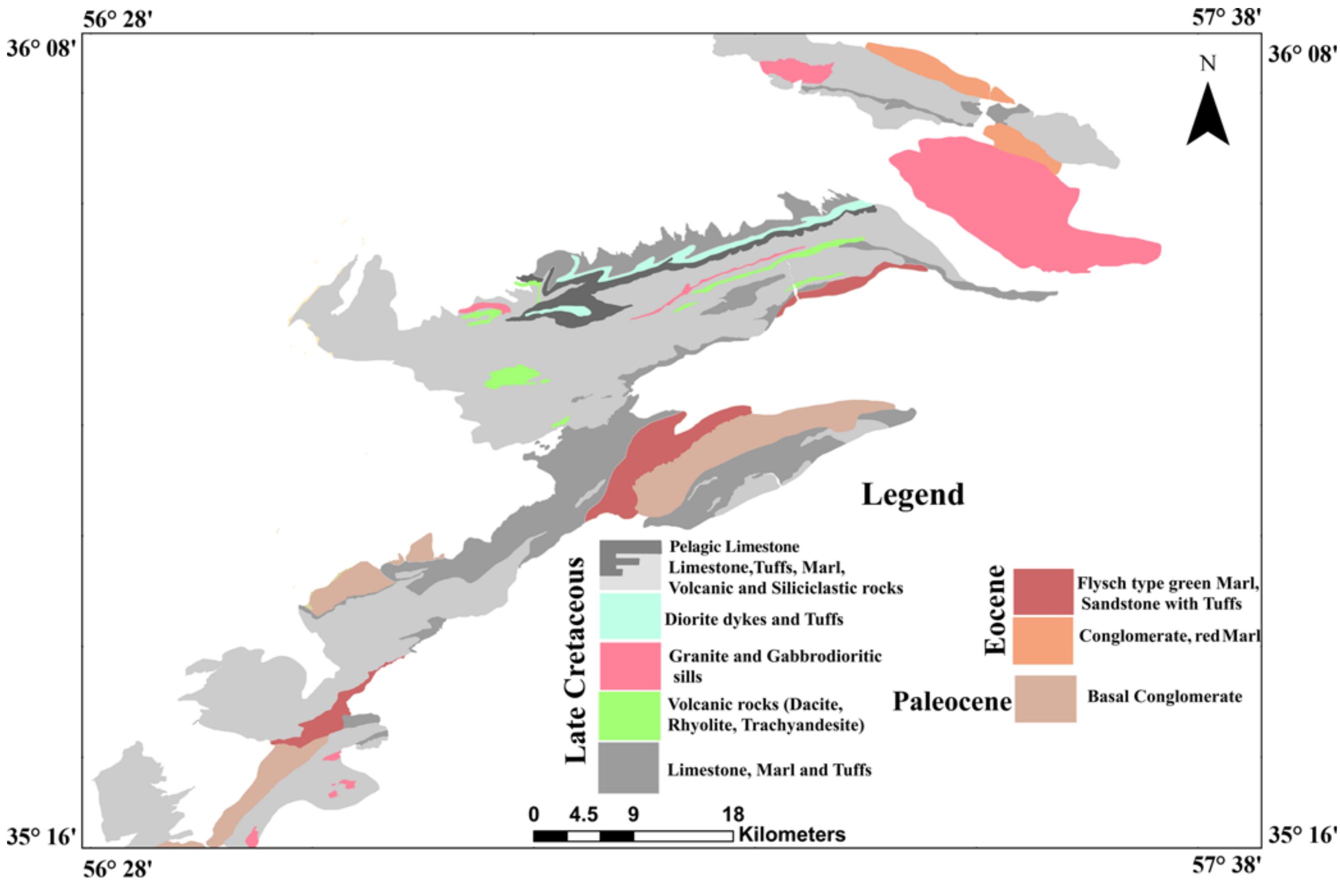

Fig. 3. Simplified geological map of the SW part of the Sabzevar zone.

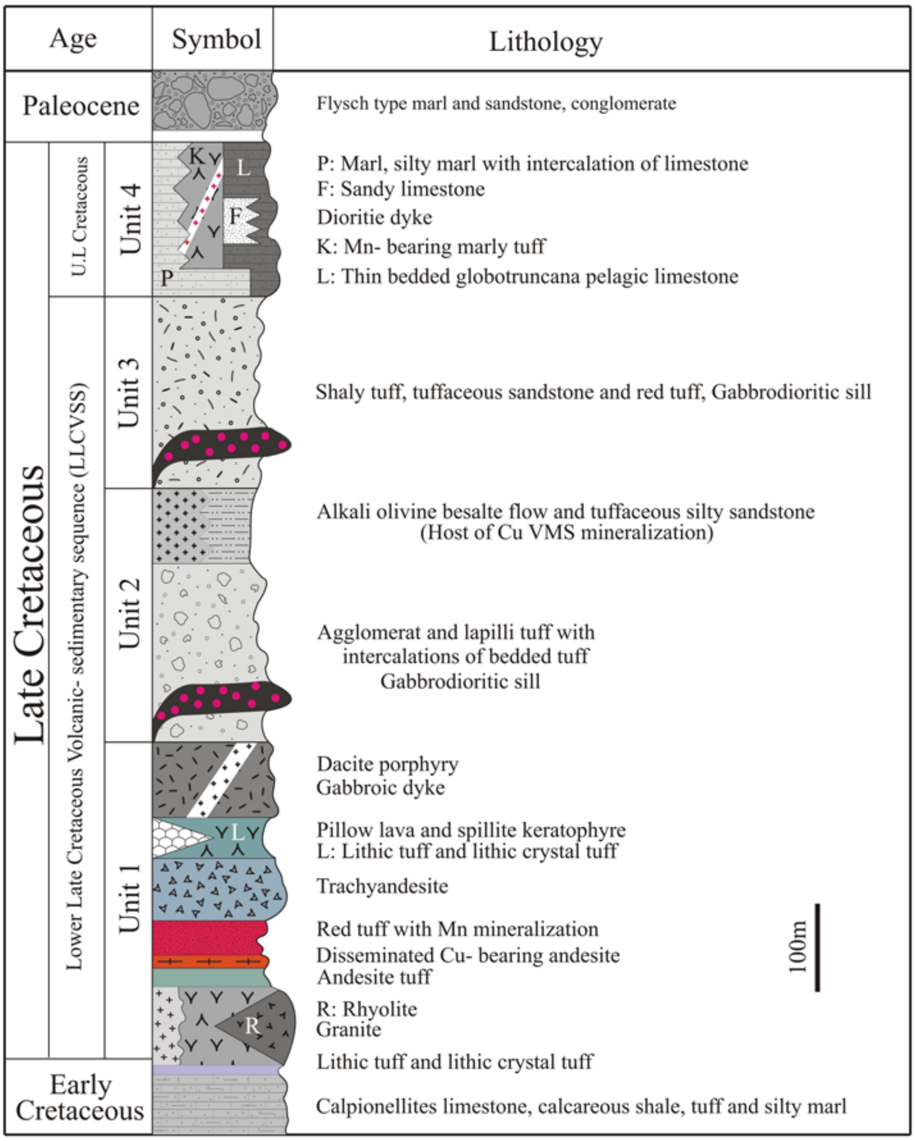

Fig. 4. Schematic geological column of the volcano-sedimentary rocks from the SW Sabzevar area (modified from Maghfouri et al. 2016).

related to the initiation of subduction along the Zagros suture zone, represented by the suprasubduction-zone-type Zagros ophiolites (Moghadam et al. 2015). Lindenberg et al. (1983) examined the structural and orogenic evolution of the Sabzevar ophiolite belt and described the area as a collection of tectonized slices on the base of the oldest nappes. Shojaat et al. (2003) identified three types of basalt with geochemical properties of normal mid-ocean ridge basalt (N-MORB), enriched MORB (E-MORB) and island arc magmas in the Sabzevar ophiolites. Moghadam et al. (2014) have recently used geochemical and geochronological data (U–Pb dating on zircons from plagiogranite) to propose that the Sabzevar ophiolite was formed 100–78 myr years ago (late Cretaceous) while the immature arc was turning into a mature arc during latest Cretaceous–Paleogene time. Blueschists, as a component of the metamorphic rocks of the Sabzevar ophiolitic complex, are related to northward-dipping subduction of the Neotethyan oceanic crust and subsequent Neotethyan closure during Early Eocene time between the Central Iranian microcontinent and Eurasia (Omrani et al. 2013).

Cenozoic magmatic rocks are abundant both north and south of the Sabzevar ophiolites and their ages range from early to late Eocene. Bauman et al. (1983) suggested that the calc-alkaline volcanic rocks from north and south of the Sabzevar ophiolites are similar to those formed in island arcs, derived by melting of the mantle wedge above an oceanic subduction zone. However, the genesis of the Paleogene igneous rocks seems to be more complex in this region. Adakites and calc-alkaline rocks are abundant among the Paleogene rocks. The genesis of adakites is suggested to be related to the fractionation of amphibole and garnet at depth whereas calc-alkaline volcanic rocks and high-K calc-alkaline granites were generated via interaction of mantle melts with the Neoproterozoic lower–middle crust of Iran and further fractional crystallization (Moghadam et al. 2015, 2016).

Regional geology

The study area is a part of the Central Iranian block, which forms the central segment of the Sabzevar zone (Pilger 1971) and is situated between the Alborz and Binalud mountains (Eurasia) in the north and the Lut block (Gondwana) in the south (Fig. 1).

The Sabzevar zone records >100 myr of magmatism and sedimentation, starting in the early Cretaceous with eruption of basaltic rocks and continuing to the Quaternary, with deposition of conglomeratic sediments and magmatic activity in the form of basaltic eruptions and/or felsic doming (Lindenberg et al. 1983). The Late Cretaceous in the Sabzevar zone is marked by back-arc opening (Sabzevar ophioites) and non-ophiolitic magmatism. Cenozoic magmatic rocks are also abundant in the Sabzevar zone and are mostly intercalated with shallow limestones (e.g. Eocene lavas) and/or terrigenous sediments (e.g. Miocene igneous rocks).

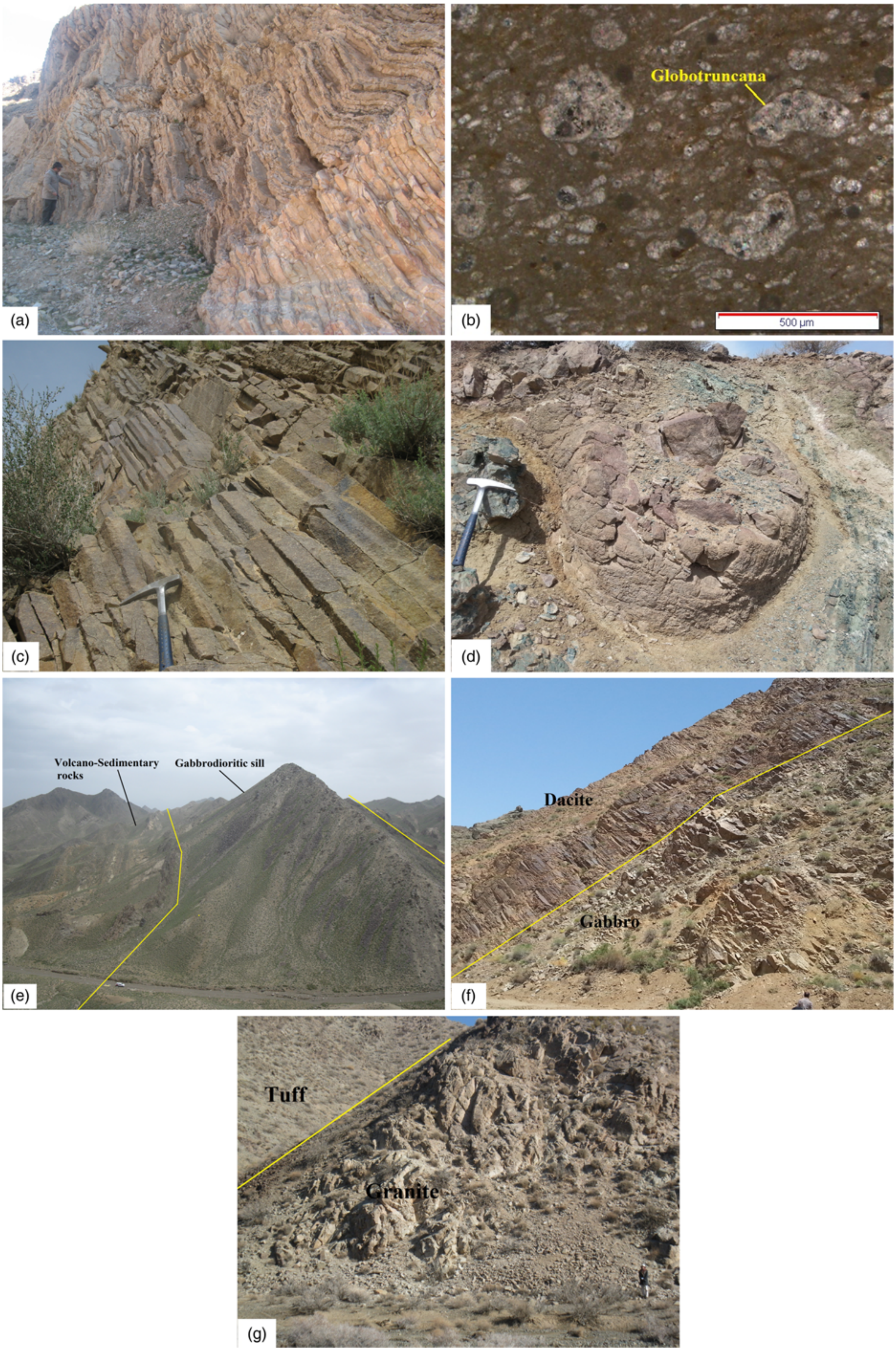

Fig. 5. Selected field views of rocks from the SW Sabzevar area. (**a**) Strongly folded *Globotruncana*-bearing pelagic limestones; (**b**) microphotograph of the *Globotruncana* limestone in plane-polarized light; (**c**) and (**d**) prismatic and exfoliation structures in the dacitic rocks; (**e**) and (**f**) gabbrodiorite sill and gabbroic dyke intruded into the Late Cretaceous volcano-sedimentary and dacitic rocks; (**g**) a view of the volcano-plutonic rocks of the study area.

Table 1. *Major (wt%) and trace element (ppm) data for the igneous samples of the SW Sabzevar area*

Sample:	Z-1	Z-2	Z-3	Z-4	Z-5	Z-6	Z-7	Z-8	Z-9	Z-10	Z-11
Rock type:	Dacite	Dacite	Rhyolite	Rhyolite	Rhyolite	Trachyandesite	Trachyandesite	Dacite	Dacite	Rhyolite	Dacite
Longitude:	57°13′10"	57°13′31"	57°11′14"	57°10′38"	57°10′35"	57°14′30"	57°14′30"	57°45′51"	57°14′09"	57°13′59"	57°29′08"
Latitude:	35°45′55"	35°45′33"	35°46′55"	35°46′38"	35°47′09"	35°41′53"	35°41′53"	35°13′19"	35°47′49"	35°48′10"	35°48′13"
SiO_2	64.08	69.93	70.71	74.93	66.85	52.09	53.56	72.60	68.65	67.85	72.13
TiO_2	0.58	0.56	0.57	0.35	0.63	0.76	0.77	0.44	0.80	0.55	0.35
Al_2O_3	13.07	13.00	12.83	11.63	12.89	16.40	16.71	11.48	13.21	12.51	12.28
$Fe_2O_3^T$	5.95	4.45	4.58	3.52	5.33	11.24	11.45	3.34	2.75	5.11	3.59
MnO	0.18	0.12	0.16	0.11	0.13	0.16	0.16	0.12	0.19	0.18	0.14
MgO	1.44	1.02	1.11	0.70	1.69	3.95	3.67	0.56	1.20	1.58	0.76
CaO	6.56	4.37	2.69	0.96	2.28	5.51	5.62	2.97	1.95	5.73	1.02
Na_2O	3.94	3.69	3.17	4.58	6.49	3.49	3.56	3.40	6.59	3.47	5.84
K_2O	1.62	0.88	0.96	1.38	0.52	4.43	3.23	1.50	0.56	0.30	0.86
P_2O_5	0.22	0.21	0.25	0.07	0.22	0.14	0.14	0.12	0.21	0.24	0.07
LOI	1.03	2.19	2.53	1.43	2.07	0.96	1.67	3.05	1.65	1.95	2.20
Sum	97.64	98.2	97.02	98.23	97.02	98.20	98.89	96.53	98.12	98.53	97.03
Be	2.35	2.49	3.17	1.42	1.73	2.64	2.57	1.59	2.59	2.45	1.62
Sc	36.832	37.20	32.21	17.05	24.30	48.74	48.81	15.86	37.23	36.41	17.51
V	118.55	106.47	26.22	29.12	97.09	385.85	381.97	78.27	53.76	112.66	29.88
Cr	18.30	18.00	5.70	9.23	9.98	22.45	22.59	7.49	9.95	6.99	7.33
Co	10.72	8.36	3.54	0.69	8.29	27.23	27.51	3.28	7.78	7.31	1.15
Ni	38.23	30.00	7.86	17.86	14.40	10.43	19.20	6.15	12.67	5.62	13.73
Cu	30.17	45.61	6.53	12.48	8.72	19.76	18.93	9.65	12.07	29.44	12.30
Zn	83.95	58.85	67.09	116.49	58.78	49.68	49.87	44.85	101.59	61.35	86.16
Ga	34.43	31.64	32.16	38.85	27.64	257.32	265.54	37.07	19.85	23.69	31.80
Rb	29.92	6.86	24.67	11.13	5.48	31.45	29.12	22.02	4.05	4.14	6.21
Sr	226.11	425.32	249.13	97.94	142.04	152.23	153.394	133.835	29.377	217.898	82.238
Y	24.06	22.44	27.59	29.15	24.06	13.56	13.68	23.14	31.86	20.29	30.45
Zr	54.83	47.09	82.43	88.45	60.66	27.34	27.21	69.89	72.95	50.74	83.91
Nb	0.92	0.77	2.03	2.13	1.55	0.50	0.51	1.28	1.27	0.76	2.18
Mo	0.78	0.85	0.45	0.72	0.77	0.39	0.38	0.76	0.68	0.56	0.74
Sn	7.55	6.41	8.81	2.94	4.91	6.12	6.27	6.37	6.91	7.98	5.44
Cs	0.52	0.16	0.21	0.03	0.05	0.45	0.54	0.30	0.07	0.10	0.00
Ba	304.71	200.92	211.71	224.82	124.07	41.34	40.32	229.31	83.05	122.26	161.85
La	5.67	6.02	5.34	4.60	5.74	4.53	4.46	5.56	4.63	5.43	4.91
Ce	12.78	12.31	14.23	13.00	14.91	9.68	9.91	12.94	12.26	11.65	13.63
Pr	2.07	1.89	1.95	2.03	2.21	1.30	1.37	1.97	1.92	1.70	2.07
Nd	9.49	9.98	8.94	10.68	10.72	6.45	6.72	9.67	10.61	8.92	10.50
Sm	2.93	2.89	3.11	3.22	3.07	1.76	1.84	2.84	3.40	2.55	3.54
Eu	1.03	1.06	1.42	1.00	0.99	0.94	0.98	0.76	1.28	1.02	1.06
Gd	3.58	3.14	4.46	4.06	3.50	2.11	2.13	3.41	4.63	3.24	4.36
Tb	0.67	0.60	0.84	0.73	0.55	0.38	0.35	0.48	0.76	0.55	0.71
Dy	4.36	3.95	4.68	5.19	4.09	2.45	2.58	3.90	5.15	3.76	5.23
Ho	0.90	0.88	1.04	1.14	0.88	0.51	0.53	0.83	1.23	0.83	1.12
Er	2.62	2.58	3.21	3.41	2.77	1.63	1.50	2.50	3.45	2.26	3.54
Tm	0.36	0.34	0.47	0.50	0.41	0.21	0.22	0.39	0.55	0.34	0.50
Yb	3.11	2.58	3.14	3.45	2.99	1.61	1.65	3.11	3.33	2.58	3.91
Lu	0.47	0.38	0.45	0.44	0.45	0.19	0.20	0.41	0.48	0.36	0.56
Hf	1.50	1.55	2.34	2.63	1.91	0.86	0.90	2.31	2.32	1.45	2.56
Ta	0.06	0.04	0.10	0.14	0.08	0.02	0.02	0.05	0.07	0.05	0.13
Pb	5.27	3.92	6.90	1.85	1.71	1.52	1.56	3.21	3.84	4.43	1.44
Th	0.93	1.04	2.07	0.79	1.28	0.71	0.76	1.26	0.56	1.01	0.84
U	0.36	0.46	0.60	0.48	0.50	0.23	0.24	0.51	0.32	0.28	0.57

Sample:	Z-12	Z-13	Z-14	Z-15	Z-16	Z-17	Z-18	Z-19	Z-20	Z-21	K-1
Rock type:	Dacite	Dacite	Rhyolite	Trachyandesite	Dacite	Dacite	Rhyolite	Dacite	Dacite	Dacite	Gabbrodiorite
Longitude:	57°28′53"	57°02′14"	57°28′54"	57°14′52"	57°02′21"	57°03′33"	57°28′31"	57°01′11"	57°03′08"	57°02′35"	57°27′46"
Latitude:	35°48′42"	35°23′26"	35°48′30"	35°42′15"	35°23′51"	35°23′49"	35°48′21"	35°23′10"	35°26′22"	35°24′11"	35°50′40"
SiO_2	66.38	65.75	67.96	67.86	61.76	68.76	71.96	70.44	71.14	72.45	53.58
TiO_2	0.61	0.62	0.55	0.77	0.68	0.55	0.40	0.71	0.48	0.47	0.64
Al_2O_3	13.26	13.23	12.27	13.68	13.22	13.56	12.79	12.07	12.73	12.14	14.95
$Fe_2O_3^T$	6.36	6.02	5.48	5.39	8.51	4.95	3.78	5.13	3.80	3.53	11.37
MnO	0.13	0.13	0.18	0.16	0.17	0.11	0.08	0.11	0.07	0.11	0.20
MgO	1.55	1.45	1.76	1.13	2.42	1.04	0.61	1.02	0.95	0.60	5.88
CaO	4.23	4.72	5.53	1.46	3.29	5.35	4.43	1.99	2.08	3.71	8.26

(continued)

Table 1. (Continued)

Sample:	Z-12	Z-13	Z-14	Z-15	Z-16	Z-17	Z-18	Z-19	Z-20	Z-21	K-1
Rock type:	Dacite	Dacite	Rhyolite	Trachyandesite	Dacite	Dacite	Rhyolite	Dacite	Dacite	Dacite	Gabbrodiorite
Na_2O	3.86	4.30	3.55	7.23	6.87	2.90	2.98	2.99	3.78	3.25	2.71
K_2O	1.50	1.60	0.46	0.09	0.14	0.66	1.19	0.66	2.36	1.67	0.79
P_2O_5	0.20	0.23	0.26	0.19	0.25	0.13	0.33	0.21	0.10	0.14	0.11
LOI	1.25	2.07	1.54	1.87	2.93	1.78	1.07	3.03	2.73	1.54	1.34
Sum	98.09	98.05	98.00	97.81	97.31	98.00	98.54	97.83	97.47	98.07	98.51
Be	2.33	2.97	2.41	2.70	2.54	1.56	1.54	2.92	3.47	1.30	1.10
Sc	37.11	43.85	35.12	36.43	39.89	27.09	27.21	32.20	28.28	11.87	42.55
V	128.59	157.43	102.26	45.37	144.95	168.73	168.56	30.18	10.47	61.66	359.60
Cr	17.54	10.11	7.81	7.88	10.47	7.87	7.67	12.05	6.25	6.93	56.52
Co	11.68	10.63	8.61	6.43	14.24	11.12	11.21	5.54	2.09	2.92	31.02
Ni	11.07	10.08	7.76	6.97	9.88	7.69	7.50	13.09	7.18	7.62	30.67
Cu	24.61	39.79	20.81	8.65	61.31	31.59	32.09	8.40	6.24	8.84	32.33
Zn	72.31	91.33	66.15	83.31	104.38	52.91	52.56	80.40	48.75	39.48	84.84
Ga	30.18	35.66	22.42	14.62	19.09	29.37	29.54	21.12	42.45	28.30	30.22
Rb	29.19	26.07	5.47	1.36	2.16	6.02	6.10	6.06	35.30	16.61	11.60
Sr	226.35	204.31	213.10	69.03	85.65	217.20	217.54	58.86	206.86	109.17	204.29
Y	20.18	25.05	22.90	28.32	24.38	23.48	23.54	34.50	29.88	15.07	13.95
Zr	45.40	51.21	46.62	71.10	53.85	52.74	84.04	78.85	29.18	43.16	34.66
Nb	0.95	1.05	0.93	1.21	1.08	0.90	0.89	1.60	2.40	0.82	0.57
Mo	0.57	0.56	0.45	0.40	0.82	0.52	0.51	0.60	0.35	0.62	0.61
Sn	5.83	7.71	8.89	6.90	9.42	2.50	2.47	0.47	8.93	2.87	2.70
Cs	0.472	0.22	0.06	0.04	0.19	0.22	0.21	0.05	0.14	0.36	0.11
Ba	235.61	294.51	136.22	28.35	36.688	143.12	143.56	130.30	307.28	150.13	126.08
La	5.12	6.05	5.76	4.37	5.47	4.51	4.45	4.90	7.70	4.32	4.00
Ce	11.86	13.55	11.71	11.34	14.11	10.18	10.10	12.66	1.65	10.08	9.19
Pr	1.85	2.17	1.94	1.82	2.26	1.74	1.78	2.16	1.92	1.52	1.18
Nd	9.11	10.25	9.39	9.70	10.43	8.86	8.65	11.71	8.85	6.90	6.55
Sm	2.71	3.14	2.97	3.40	3.46	2.73	2.64	3.87	3.21	2.11	1.94
Eu	0.98	1.09	0.89	1.19	1.13	0.84	0.81	1.12	1.41	0.78	0.68
Gd	3.29	3.69	3.48	4.24	3.32	3.24	3.27	4.87	4.51	2.33	2.05
Tb	0.54	0.65	0.56	0.72	0.52	0.59	0.58	0.91	0.83	0.44	0.37
Dy	3.74	4.37	3.95	5.37	4.28	4.38	4.29	6.33	5.57	2.74	2.53
Ho	0.72	0.96	0.86	1.17	0.92	0.93	0.91	1.33	1.04	0.71	0.55
Er	2.35	2.83	2.69	3.22	2.91	2.64	2.49	4.17	3.45	1.71	1.53
Tm	0.31	0.45	0.39	0.47	0.47	0.39	0.38	0.57	0.54	0.24	0.15
Yb	2.28	2.88	2.47	3.28	3.11	2.79	2.70	3.91	4.00	1.93	1.70
Lu	0.32	0.40	0.38	0.47	0.46	0.40	0.38	0.58	0.55	0.27	0.28
Hf	1.40	1.74	1.49	2.29	1.77	1.65	1.39	2.70	3.43	1.46	1.04
Ta	0.06	0.05	0.04	0.06	0.06	0.06	0.06	0.10	0.08	27.19	0.08
Pb	4.19	6.42	3.57	2.37	7.20	4.49	4.56	3.78	2.31	0.97	5.60
Th	1.08	1.15	1.07	0.61	1.14	0.78	0.77	0.67	2.99	0.49	0.80
U	0.28	0.37	0.23	0.30	0.45	0.39	0.37	0.50	0.75	27.19	0.28

Sample:	K-2	K-3	K-4	K-5	K-6	K-7	K-8	K-9	K-10	K-11	K-12
Rock type:	Diorite	Gabbrodiorite	Gabbrodiorite	Gabbrodiorite	Gabbro	Granite	Granite	Granite	Granite	Granite	Granite
Longitude:	57°21′05"	57°27′54"	57°28′14"	57°28′23"	57°28′40"	57°02′15"	57°02′31"	57°03′2"	57°03′47"	57°02′57"	57°04′11"
Latitude:	35°49′38"	35°50′34"	35°50′04"	35°49′36"	35°49′12"	35°12′18"	35°26′58"	35°27′0"	35°26′25"	35°26′10"	35°26′31"
SiO_2	53.34	59.99	53.84	68.96	52.45	75.01	74.84	77.95	73.84	78.00	77.12
TiO_2	0.49	0.80	0.66	0.55	0.75	0.24	0.22	0.20	0.40	0.17	0.26
Al_2O_3	17.80	15.35	16.43	12.27	18.20	12.92	12.78	11.20	12.01	11.53	11.85
$Fe_2O_3^T$	8.31	9.01	9.12	5.48	9.82	1.25	2.30	1.40	3.60	1.90	1.51
MnO	0.19	0.19	0.14	0.18	0.17	0.03	0.02	0.02	0.07	0.01	0.05
MgO	3.31	2.16	2.56	1.76	2.75	0.81	0.70	0.63	1.28	0.74	0.63
CaO	7.95	6.21	9.59	5.53	6.33	2.78	2.18	1.93	2.73	0.74	2.07
Na_2O	3.52	3.32	2.95	3.55	4.44	5.03	4.70	4.60	3.60	5.31	4.35
K_2O	0.77	1.32	0.63	0.46	2.30	0.30	0.31	0.23	0.34	0.19	0.26
P_2O_5	0.12	0.22	0.13	0.26	0.15	0.02	0.02	0.02	0.07	0.02	0.03
LOI	3.12	2.34	2.65	1.24	2.14	3.04	2.76	1.64	2.61	2.35	1.89
Sum	95.8	98.60	96.08	99.00	97.41	98.41	98.07	98.1	97.95	98.60	98.1
Be	1.46	1.44	1.58	2.26	2.06	1.70	1.62	1.56	1.71	1.46	1.49
Sc	19.41	27.65	26.43	24.10	39.61	16.79	13.02	12.05	18.76	12.52	13.61
V	231.1	213.33	324.20	166.17	317.41	54.23	45.16	43.18	64.70	39.04	35.79
Cr	9.25	11.46	7.45	9.85	6.26	7.42	3.82	2.78	4.79	6.56	8.85

(continued)

Table 1. (Continued)

Sample:	Z-12	Z-13	Z-14	Z-15	Z-16	Z-17	Z-18	Z-19	Z-20	Z-21	K-1
Rock type:	Dacite	Dacite	Rhyolite	Trachyandesite	Dacite	Dacite	Rhyolite	Dacite	Dacite	Dacite	Gabbrodiorite
Co	16.05	16.22	19.26	15.57	21.05	1.57	1.93	1.85	5.32	5.86	3.25
Ni	11.49	13.44	13.04	9.72	11.90	14.14	7.18	6.56	6.66	6.69	12.19
Cu	34.61	44.81	179.09	28.56	210.00	6.62	5.91	4.78	11.05	5.40	9.08
Zn	61.46	107.31	61.45	59.93	83.47	5.96	7.94	6.65	13.47	9.58	17.91
Ga	46.24	47.33	31.23	32.24	40.56	16.52	13.59	12.97	16.40	9.78	17.40
Rb	11.41	21.41	10.10	17.38	29.92	1.80	1.91	1.67	2.53	1.09	1.92
Sr	541.15	233.51	340.34	350.54	329.84	134.78	105.18	103.71	122.23	72.19	105.07
Y	10.20	19.92	12.64	9.24	13.86	40.96	43.66	42.97	29.08	26.52	19.26
Zr	49.91	47.88	32.07	46.00	33.30	118.08	125.92	123.87	63.13	63.22	92.18
Nb	2.12	1.01	0.69	2.30	0.82	3.35	2.70	1.98	1.23	1.32	1.80
Mo	0.68	0.85	0.47	0.21	0.98	0.20	0.33	0.26	0.36	1.11	0.37
Sn	2.58	3.53	2.66	6.81	6.15	2.04	9.94	8.65	2.30	3.34	6.43
Cs	0.06	0.24	0.36	0.706	0.54	0.09	0.25	0.17	0.17	0.02	0.13
Ba	227.27	243.62	125.78	233.42	280.97	59.24	40.15	39.00	46.61	20.14	53.91
La	6.53	5.97	4.47	6.29	4.16	4.20	7.24	6.43	3.48	2.61	7.91
Ce	18.51	14.10	9.48	13.11	9.15	12.92	17..28	15.97	8.71	6.88	15.66
Pr	2.04	1.81	1.02	1.74	1.32	2.03	2.21	2.03	1.34	1.05	1.58
Nd	9.18	9.77	6.34	7.04	6.02	10.80	11.77	10.34	7.49	6.09	6.77
Sm	1.80	2.73	1.85	2.19	2.00	3.66	4.13	3.43	2.35	2.34	1.51
Eu	0.52	0.90	0.71	0.61	0.73	0.53	0.47	0.37	0.66	0.42	0.63
Gd	1.89	3.12	1.79	1.56	2.07	4.87	5.11	4.04	3.33	2.71	2.12
Tb	0.42	0.60	0.37	0.27	0.33	0.82	1.00	0.87	0.63	0.56	0.36
Dy	3.23	3.52	2.49	1.70	2.39	6.33	7.23	6.43	4.74	4.88	2.66
Ho	0.65	0.68	0.57	0.40	0.53	1.48	1.67	1.43	1.04	1.02	0.64
Er	1.89	2.21	1.48	0.99	1.28	4.76	4.92	3.65	3.25	3.30	2.00
Tm	0.28	0.23	0.20	0.16	0.20	0.67	0.74	0.69	0.49	0.53	0.31
Yb	2.10	2.69	1.59	1.20	1.59	5.40	5.85	4.67	3.71	3.41	2.68
Lu	0.31	0.35	0.23	0.18	0.25	0.80	0.86	0.76	0.49	0.56	0.41
Hf	1.37	1.34	1.02	1.26	1.22	3.83	4.11	3.23	2.06	2.04	3.02
Ta	0.07	0.07	0.02	0.09	0.04	0.21	0.13	0.12	0.10	0.07	3.02
Pb	2.31	6.11	1.50	3.12	3.24	1.12	1.70	1.60	1.26	0.94	1.54
Th	2.19	1.12	0.80	1.77	0.91	1.95	2.26	1.98	0.60	0.69	1.07
U	0.73	0.24	0.27	0.49	0.29	0.72	0.93	0.86	0.38	0.41	0.56

Previous studies in the Sabzevar zone have focused on the general geology, petrography and petrology of the Late Cretaceous Cu, Zn and Mn mineralization and their host rocks (Maghfouri *et al.* 2016). The stratigraphic section of the non-ophiolitic late Cretaceous rocks from the Sabzevar zone is dominated by igneous rocks with interbedded sedimentary and volcanoclastic rocks (Fig. 3). All

Table 2. *Results of electron microprobe analysis of plagioclases*

Sample	SiO_2	TiO_2	Al_2O_3	FeO	MnO	MgO	CaO	Na_2O	K_2O	Total
Z-18-01	49.85	0.008	31.17	0.621	0.021	0.045	14.68	3.21	0.04	99.76
Z-18-02	49.54	0.000	31.57	0.644	0.026	0.048	1504	2.98	0.03	99.95
Z-18-03	50.69	0.028	30.71	0.613	0.000	0.040	14.25	3.43	0.03	99.80
Z-18-04	51.18	0.000	30.41	0.619	0.001	0.053	13.93	3.61	0.04	99.95
Z-3-01	49.98	0.028	31.27	0.641	0.003	0.036	11.27	2.64	0.02	99.90
Z-3-02	49.22	0.045	31.30	0.717	0.013	0.053	15.31	2.77	0.05	99.50
Z-3-03	68.21	0.000	19.61	0.023	0.000	0.010	0.08	11.84	0.01	99.77
Z-3-04	67.04	0.006	19.73	0.000	0.000	0.015	0.27	11.26	0.11	98.45
Z-17-01	67.19	0.022	20.73	0.050	0.000	0.000	0.71	11.36	0.07	100.17
Z-17-02	68.39	0.000	19.92	1.062	0.023	0.458	0.41	11.14	0.12	101.59
Z-17-03	67.59	0.000	21.12	0.246	0.013	0.044	0.60	10.71	0.53	100.87
Z-17-04	67.44	0.033	20.68	0.062	0.006	0.000	0.43	11.65	0.11	100.48
K-1-01	48.59	0.000	32.26	0.859	0.018	0.101	16.06	2.58	0.06	100.55
K-1-02	46.44	0.028	33.37	0.896	0.000	0.074	17.50	1.69	0.02	100.10
K-1-03	47.55	0.045	32.73	0.887	0.000	0.121	16.72	2.16	0.02	100.24
K-1-04	46.62	0.037	32.98	0.917	0.000	0.085	17.12	1.86	0.03	99.69
K-10-01	60.89	0.005	24.54	0.479	0.015	0.074	5.76	8.25	0.44	100.48
K-10-02	60.03	0.000	24.94	0.247	0.005	0.001	6.28	8.01	0.44	100.10
K-10-03	58.04	0.000	26.67	0.341	0.018	0.039	8.36	6.88	0.29	100.67
K-10-04	59.80	0.008	25.07	0.784	0.000	0.068	6.61	7.71	0.41	100.47

Table 3. *Results of electron microprobe analysis of clinopyroxenes*

Sample	SiO_2	TiO_2	Al_2O_3	FeO	MnO	MgO	CaO	Na_2O	K_2O	Cr_2O_3	Total
Z-18-01	52.07	0.35	1.44	11.52	0.58	13.91	20.23	0.25	0.01	0.00	100.36
Z-18-02	51.31	0.46	1.73	11.78	0.59	13.76	20.33	0.27	0.00	0.00	100.34
Z-18-03	51.85	0.25	1.32	11.78	0.66	14.11	20.03	0.25	0.01	0.00	100.26
Z-18-04	51.79	0.25	1.41	11.60	0.58	13.90	20.15	0.26	0.00	0.02	99.97
Z-3-01	51.20	0.54	1.57	13.37	0.62	12.77	18.80	0.22	0.00	0.01	99.10
Z-3-02	52.20	0.26	1.22	12.73	0.72	13.36	19.04	0.26	0.00	0.00	99.85
Z-3-03	52.52	0.36	1.45	12.83	0.73	13.20	18.56	0.26	0.00	0.00	99.92
Z-3-04	52.30	0.28	1.58	11.96	0.67	13.83	19.50	0.25	0.00	0.00	100.35
K-1-01	51.68	0.38	1.50	13.00	0.59	14.13	18.99	0.23	0.00	0.00	100.58
K-1-02	51.39	0.37	1.53	13.46	0.54	13.74	19.04	0.23	0.00	0.00	100.32
K-1-03	51.07	0.42	1.88	13.40	0.55	13.28	19.36	0.23	0.01	0.02	100.23
K-1-04	52.17	0.40	2.35	10.35	0.27	15.31	19.75	0.24	0.00	0.01	100.89

volcano-sedimentary units have been intruded by dacite, rhyolite, shallow granitoid and gabbroic bodies. The generalized volcano-sedimentary sequence of the SW Sabzevar area consists of four major units based on their stratigraphic and compositional variations and age dating (Fig. 4). In bottom-up order, unit 1 is identified by its abundant igneous rocks, such as dacite, rhyolite, trachyandesite and granite along with lithic tuff and crystal lithic tuff. The second unit of the sequence comprises abundant agglomerate, lapilli tuff and a minor amount of tuffaceous sandstone. Unit 3 mainly consists of pyroclastic rocks more than 700 m thick, cut by gabbrodiorite and gabbro sills. The uppermost unit (unit 4) of the sequence is mostly composed of *Globotruncana*-bearing pelagic limestones along with tuffaceous marls, silty marls and sandy limestones, which have been cut by diorite dykes in some places. The index fossils in the limestones indicate a Late Cretaceous age for the volcano-sedimentary sequence (see also Lindenberg *et al.* 1983; Maghfouri *et al.* 2016; Fig. 5a and b). The facies diversity of the rocks reflects the different sedimentary environments that are especially found in island arc settings (McPhie *et al.* 1993). The dacitic unit, the thickest felsic volcanic unit, is distributed mainly in Kal-e Morgh Valley and north of Nudeh village, with rough prominent topography, columnar jointing, and prismatic and exfoliated structures (Fig. 5c and d).

There are concurrent gabbrodiorite sub-volcanic intrusions within the volcano-sedimentary units of the region. On the basis of geological, mineralogical and textural characteristics, we have subdivided the plutonic rocks into three types: diorite, gabbro-diorite and granite. These rocks have intruded as stocks, sills and dykes into the volcanic and volcano-sedimentary rocks of the Late Cretaceous sequence and crop out in the SW of the Sabzevar zone (Fig. 5e). The gabbro dykes also intruded the dacitic rocks (Fig. 5f). Granitic rocks NE of Asbkeshan village have intruded as large and small stocks into the Late Cretaceous volcano-sedimentary units and caused chloritic and argillitic alteration near the contact (Fig. 5g). Granitic rocks contain rounded microgranular enclaves of diorite, which are darker and have sharp boundaries with the host rock.

Analytical methods

Whole-rock geochemistry, analyses of radiogenic isotopes and zircon U–Pb dating were performed in the Geochemical Analysis Unit of the ARC Centre of Excellence for Core to Crust Fluid Systems (CCFS/GEMOC) Macquarie University, Australia. Electron microprobe analyses were performed at the University of New South Wales, Australia.

Thirty-three samples with the least alteration were selected from the Late Cretaceous volcanic and intrusive rocks for whole-rock major- and trace-element analyses. Whole-rock major elements were analysed in fused lithium tetraborate ($LiBO_2$) glass beads using an X-ray fluorescence (XRF) spectrometer. Loss on ignition (LOI) was obtained based on the mass difference of 1 g of dried powdered samples before and after heating for 1 h at 1000°C. Rare earth elements (REE) and other trace elements were determined by laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) on fused glass discs using $Li_2B_4O_7$ (lithium metaborate flux). Major- and trace-element data for both the volcanic and intrusive rocks are listed in Table 1. The international standards BCR2-G, Nist 610 and BHVO-1 were used for calibration (Norman *et al.* 2006). Detailed analytical methods for the whole-rock analysis have been presented by Neumann *et al.* (2004).

Major-element compositions of the clinopyroxene, plagioclase and amphibole and backscattered electron (BSE) images were obtained at the Australian Microscopy and Microanalysis Research Facility (AMMRF), University of New South Wales with a JEOL JXA-8100 Superprobe electron microprobe (EMP) paired with an Oxford Instruments INCA energy-dispersive spectrometry (EDS) system using an electron beam with current of 20 nA and accelerating voltage of 15 kV focused to a spot 5 µm in diameter. Analyses were made on cores, rims and intermediate points of the mineral phases to assess compositional zonation. Analytical precision (2σ error), as evaluated by repeated analyses of single grains, is better than 1 to 5% for elements in concentrations of >20 to 2 wt% oxide respectively, and better than 10% for elements in the range 0.5–2 wt% oxide. Full results are given in Tables 2–4.

Table 4. *Results of electron microprobe analysis of amphiboles*

Sample	SiO_2	TiO_2	Al_2O_3	FeO	MnO	MgO	CaO	Na_2O	K_2O	Total
K-2-01	48.913	1.021	9.845	14.898	0.663	12.78	11.316	1.55	0.258	100.31
K-2-02	47.197	1.11	9.478	14.755	0.701	13.063	11.156	1.568	0.232	99.34
K-2-03	46.891	1.049	9.789	15.102	0.652	12.973	11.242	1.562	0.231	99.56
K-2-04	48.672	0.996	8.324	14.211	0.655	13.572	11.307	1.375	0.176	99.31
K-10-01	50.683	0.293	3.084	23.483	0.827	9.969	9.639	0.823	0.235	99.38
K-10-02	51.331	0.237	3.424	23.465	0.754	9.667	9.98	0.898	0.266	100.38
K-10-03	49.788	0.238	3.845	23.498	0.66	9.67	10.182	0.989	0.292	99.59
K-10-04	47.351	1.243	5.6	21.441	0.782	10.303	10.15	1.738	0.46	99.50

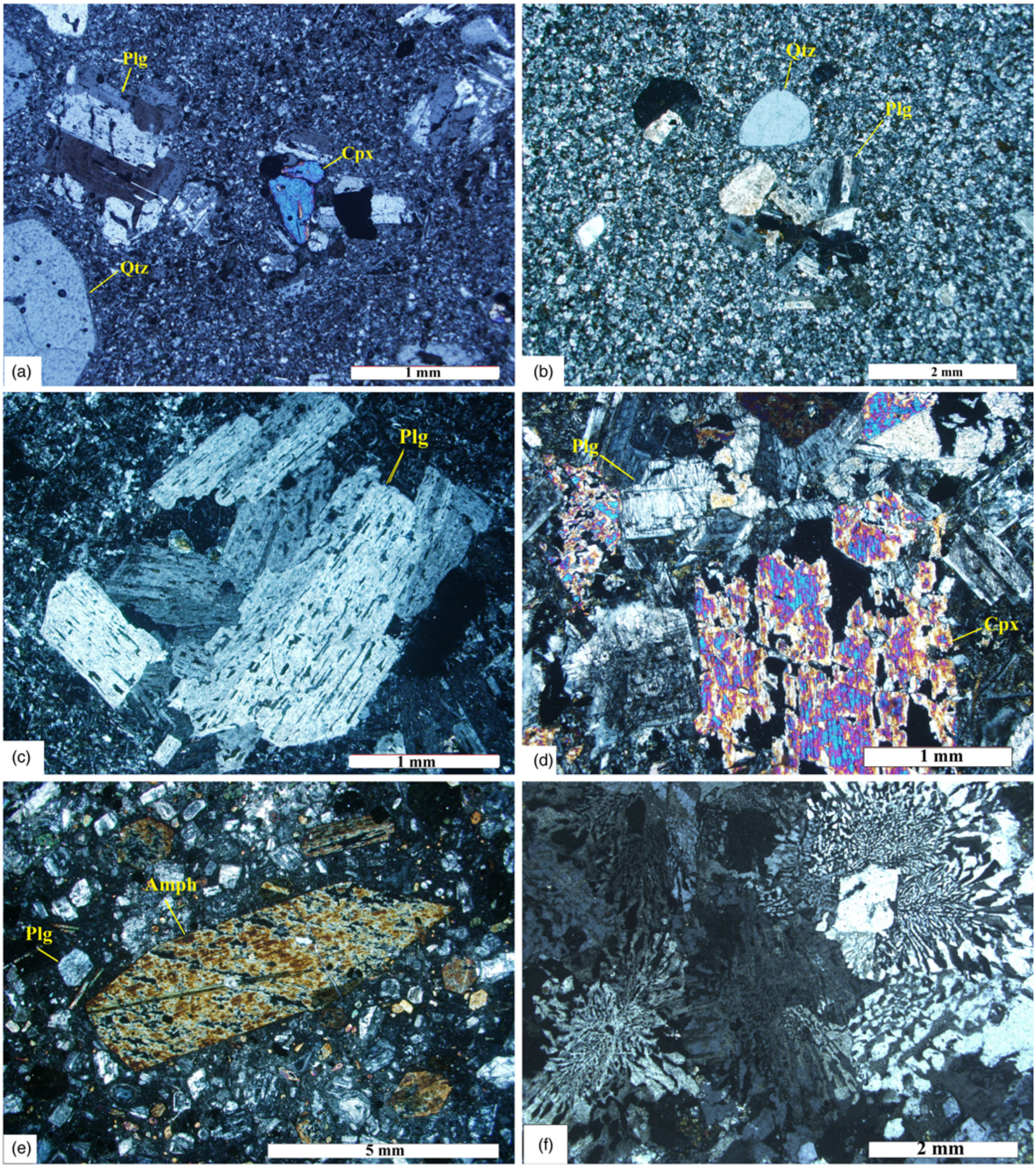

Fig. 6. Photomicrographs showing the mineralogy and textures of the SW Sabzevar Late Cretaceous extrusive and intrusive rocks. (**a**) Felsitic glomeroporphyritic texture composed of quartz, plagioclase and clinopyroxene phenocrysts in a dacitic sample (cross-polarized light; XPL); (**b**) plagioclase and quartz phenocrysts in the fine-grained plagioclase–quartz matrix of a felsic porphyritic rhyolite (XPL); (**c**) a glomeroblast of plagioclase crystals with sieve texture in a trachyandesite (XPL); (**d**) phenocrysts of clinopyroxene and plagioclase in a gabbrodiorite sill (XPL); (**e**) a euhedral amphibole crystal ina diorite porphyry; (**f**) granophyric texture in an alkali granite sample (XPL). Abbreviations: amph, amphibole; cpx, clinopyroxene; plg, plagioclase; qtz, quartz.

Zircons were separated from five samples; single grains were hand-picked under a binocular microscope and mounted in epoxy discs. Zircon U–Pb ages were obtained using a 193 nm ArF excimer laser with an Agilent 7700 ICP-MS system. Detailed method descriptions have been given by Jackson *et al.* (2004). Prior to analysis single zircon grains were imaged and characterized using a Merchantek LUV 266 scanning electron microscope (SEM) to determine the presence or absence of multiple growth phases, xenocrystic cores, inclusions and cracks. Imaging was carried out in both secondary electron (SE) and BSE modes, and suitable, representative zircons and specific locations for analysis were identified. The ablation conditions included beam size (30 µm), pulse rate (5 Hz) and energy density (7.59 J cm^{-2}). Analytical runs comprised 16 analyses with 12 analyses of

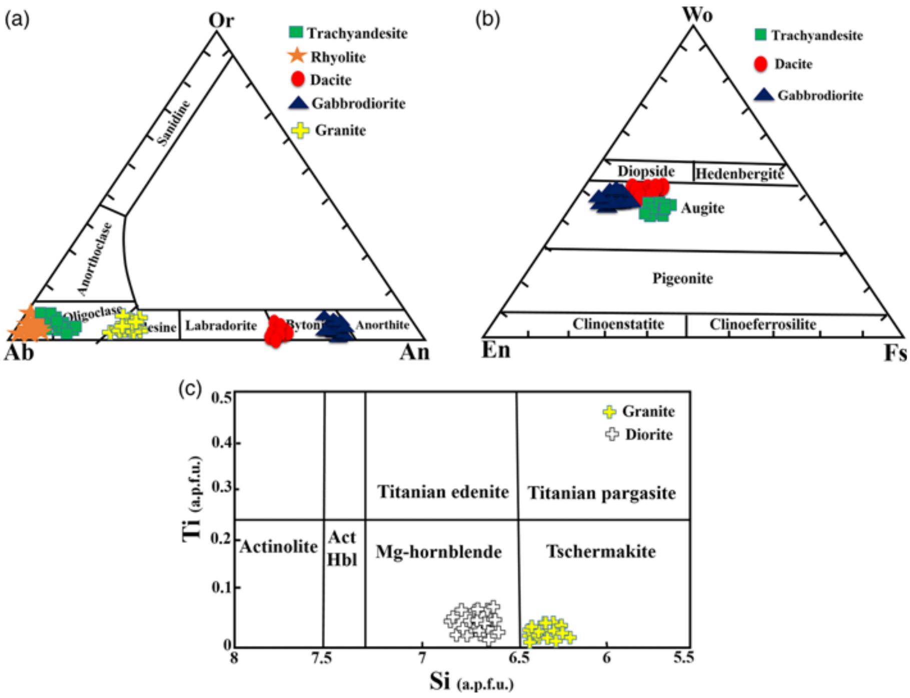

Fig. 7. Plot of feldspar and clinopyroxene compositions of the Sabzevar intrusive and extrusive rocks in (**a**) ternary An–Or–Ab diagram (after Deer *et al.* 1966), (**b**) clinopyroxene classification diagram (after Morimoto *et al.* 1988) and (**c**) amphibole classification diagram (Cao *et al.* 2017) for granitic and dioritic rocks.

unknowns bracketed by two analyses of a standard zircon GJ-1 at the beginning and end of each run, using the established thermal ionization mass spectrometry (TIMS) values ($^{207}Pb/^{206}Pb$ age = 608.5 Ma) (Jackson *et al.* 2004). U–Pb ages were calculated from the raw signal data using the on-line software package GLITTER (Griffin *et al.* 2008; WWW.mq.edu.au/GEMOC). GLITTER calculates the relevant isotopic ratios for each mass sweep and displays them as time-resolved data. This allows isotopically homogeneous segments of the signal to be selected for integration. GLITTER then corrects the integrated ratios for ablation-related fractionation and instrumental mass bias by calibration of each selected time-segment against the identical time-segments of the standard zircon analyses (standard GJ-1).

U–Pb age data were subjected to a common-lead correction, except for those with common-Pb concentrations lower than detection limits. We have followed the correction procedure of Andersen (2002), which assumes that the observed $^{206}Pb/^{238}U$, $^{207}Pb/^{235}U$ and $^{208}Pb/^{232}Th$ ratios of a discordant zircon reflect a combination of common lead and lead loss at a defined time and solves the resulting mass-balance equations. The amount of common lead and the common-lead-corrected composition of the sample are determined simultaneously. This procedure yields common-lead corrections of similar magnitude to those obtained from measurements of ^{204}Pb in ion-probe analysis; the relative error in the amount of common lead is <10%. The uncertainty in common-lead concentration contributes significantly to the uncertainty in the $^{207}Pb/^{206}Pb$ age only above 0.5% common lead. The analyses presented here have been corrected assuming recent lead-loss and a common-lead composition corresponding to present-day average orogenic lead as given by the second-stage growth curve of Stacey & Kramers (1975) for $^{238}U/^{204}Pb = 9.74$. No correction has been applied to analyses that are concordant within 2σ analytical errors in $^{206}Pb/^{238}U$ and $^{207}Pb/^{235}U$, or that have <0.2% common lead. The results were processed using the ISOPLOT program of Ludwig (2003). The external standards, zircons 91500 and Mud Tank, gave mean $^{206}Pb/^{238}U$ ages of 1063.5 ± 1.8 Ma (MSWD = 1.3) and 731.1 ± 1.2 Ma (MSWD = 0.77), respectively, which are similar to the recommended $^{206}Pb/^{238}U$ ages of 1062.4 ± 0.4 Ma and 731.9 ± 3.4 Ma respectively (Woodhead & Hergt 2005; Chang *et al.* 2006; Yuan *et al.* 2008). More details about the zircon U–Pb dating have been presented by Moghadam *et al.* (2017). The zircon U-Pb isotopic data are summarised in Tables 5–7.

Fourteen samples of fresh dacite, rhyolite, diorite and granite from the SW Sabzevar area were selected for determination of Rb–Sr and Sm–Nd isotopic ratios. Samples were digested in 15 ml Savillex Teflon beakers using concentrated HF (Merck Suprapur) and Teflon-distilled concentrated HNO_3 (Merck AR grade). Samples were then dried down and 10 drops of concentrated $HClO_4$ (Merck Suprapur) were added, followed by 4 ml 6N HCl and H_2O_2. Samples were dried down again, dissolved in 6N HCl, dried, dissolved in 2.5N HCl–0.1N HF, and centrifuged to remove any undissolved residue. Samples were then loaded onto Teflon columns containing Biorad® AG50W-X8 (200–400 mesh) cation exchange resin and eluted using a 2.5N HCl–0.1N HF solution. Hf and matrix elements was eluted with 5.4 ml of this solution. Strontium was collected after 34.9 ml of 2.5N HCl, followed by Nd. Neodymium in 6N HCl was separated from Sm, Ba, La and Ce using a second column (Eichrom® Ln spec resin (50–100 µm); e.g. Aulbach *et al.* 2004). Isotopic analyses of Sr and Nd were obtained by TIMS using a Thermo Finnigan Triton system. Samples for Sr isotopic analysis were loaded onto single rhenium filaments with a Ta activator. Analyses were made between 1380 and 1430°C, with total Sr signals of 1–11 V; they were collected in static mode with amplifier rotation and were collected in 10 blocks of 20 cycles each, with an integration time of 4.2 s per cycle. Ratios were normalized to $^{86}Sr/^{88}Sr = 0.1194$ to correct for mass fractionation. Measured $^{87}Sr/^{86}Sr$ ratios for BHVO-2 (average = 0.703457 ± 3, $n = 2$) and NIST SRM 987 (average = 0.710208 ± 3, $n = 2$) are in good agreement with reference values. Samples for Nd isotopic analysis were loaded onto double rhenium filaments and analysed with an evaporation filament current of 1200–1600 mA and a total Nd signal of 0.5–10 V. Analyses were collected in static mode with amplifier rotation, collected in four blocks of 50 cycles each, with an integration time of 8.4 s per cycle. Ratios were normalized to $^{146}Nd/^{144}Nd = 0.7219$ to correct for mass fractionation. Measured values for standards BHVO-2 (average = 0.512987 ± 21, $n = 4$) and NIST JMC 321 (0.511118 ± 5, $n = 1$) are in good agreement with reference values and are presented in Table 8. Detailed analytical methods for whole-rock analysis have been presented by Neumann *et al.* (2004).

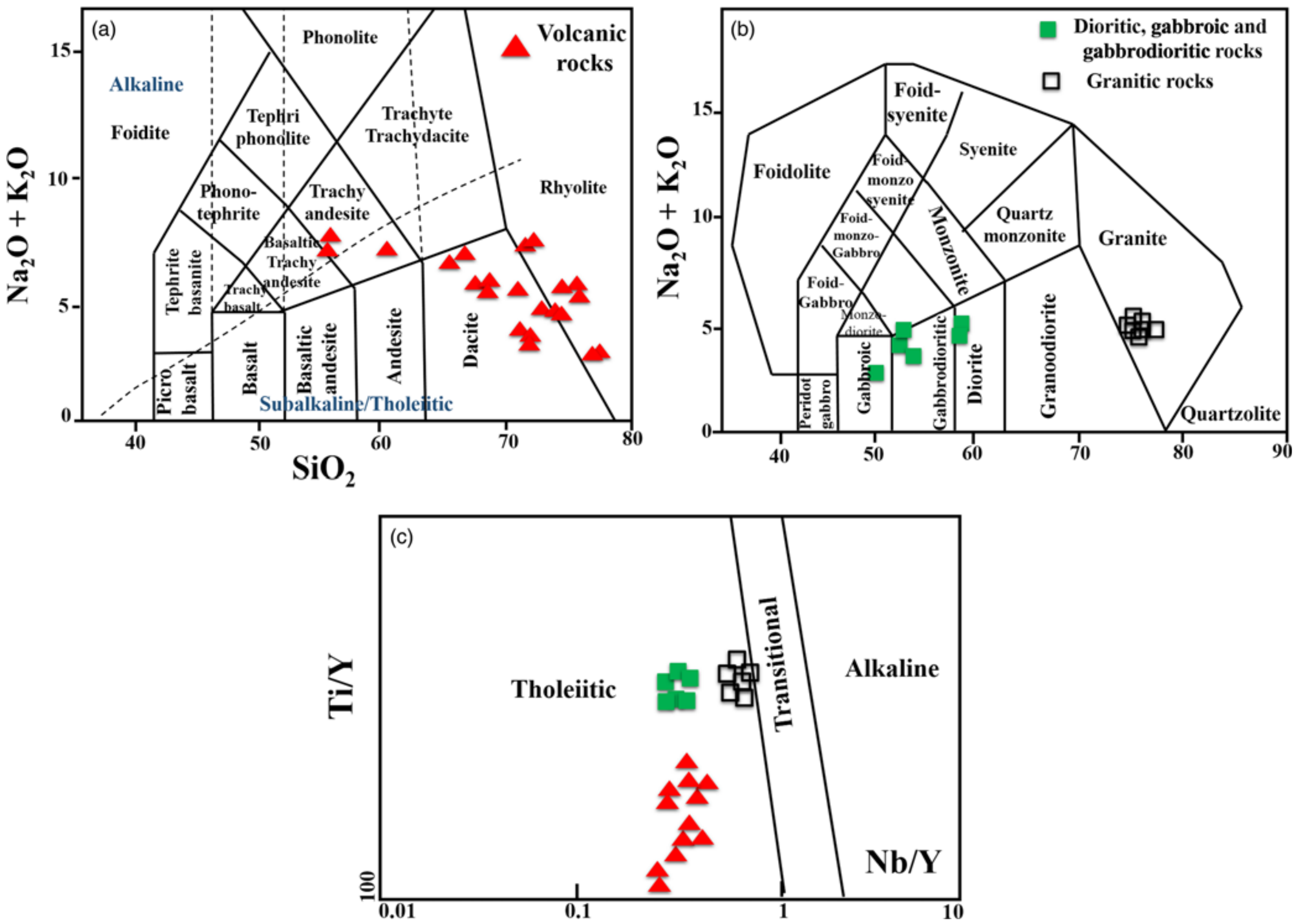

Fig. 8. Plots of the Late Cretaceous intrusive and extrusive rocks from the SW Sabzevar area in different discriminative diagrams: (**a**) $Na_2O + K_2O$–SiO_2 (Le Bas *et al.* 1986); (**b**) $Na_2O + K_2O$–SiO_2 (Middlemost 1985); (**c**) Nb/Y–Ti/Y (Pearce 1982), showing the nature of magma.

Results

Petrography

Microscopic observations indicate that the fresh dacitic rocks are composed of 25–35 vol% phenocrysts, chiefly plagioclase (10–15 vol%), quartz (8–12 vol%) and clinopyroxene (5–8 vol%) (Fig. 6a). A porphyritic texture is typical in the dacitic rocks, which show phenocrysts of plagioclase and clinopyroxene embedded in a microcrystalline groundmass of plagioclase and quartz. Plagioclase occurs as subhedral to euhedral laths, which have sieve textures that may be a result of rapid decompression (Nelson & Montana 1992). Some of the large plagioclase phenocrysts in the fresh dacites exhibit simple twinning and normal zoning. Usually quartz phenocrysts in dacitic samples have embayed surfaces and rounded irregular shapes and are up to 2–5 mm in size. Clinopyroxene grains in dacites are subhedral to euhedral and 1 mm in maximum length; they are usually surrounded by an irregular corrosion rim and sometimes have plagioclase inclusions. Plagioclase and quartz are the dominant mineral phases in the rhyolitic rocks (10–15 vol%), and phenocrysts of each are embedded in a groundmass of plagioclase and quartz (Fig. 6b). Plagioclase phenocrysts have euhedral to subhedral shapes and are up to 2 mm in size. Some of the plagioclase shows varying degrees of alteration to sericite, chlorite and clays. Quartz phenocrysts usually occur as irregular crystals and they have embayed surfaces, suggesting resorption by the magma.

The trachyandesites are composed of plagioclase and clinopyroxene phenocrysts in a groundmass of plagioclase microlites, glass, quartz and minute grains of clinopyroxene. Plagioclase phenocrysts (10–20 vol%) occur as euhedral to subhedral grains in porphyritic and glomeroporphyritic textures. Plagioclase grains often show sieve texture and are partly resorbed (Fig. 6c). Clinopyroxene grains form 5–8 vol% of the trachyandesites.

Gabbrodiorites are coarse- to medium-grained and are composed of plagioclase, clinopyroxene and opaque minerals with minor amounts of amphibole and quartz (Fig. 6d). The plagioclase occurs as large euhedral to subhedral twinned prisms as well as small prisms enclosed in clinopyroxene and shows various degrees of alteration to sericite and chlorite. Clinopyroxene is generally euhedral to subhedral and forms unzoned grains up to 2 mm in size. Some clinopyroxene crystals contain numerous tiny inclusions of opaque oxides and plagioclase, which are altered. Small euhedral grains of apatite occur as inclusions in plagioclase.

The dioritic rocks are composed of plagioclase (45–50 vol%), amphibole (30–40 vol%) and opaque minerals (8–10 vol%) in a fine-grained groundmass. Amphibole crystals commonly have rhombic and prismatic forms and are multi-coloured green to brown in the dioritic rocks. Their size varies from 0.5 to 10 mm, and they often contain inclusions of plagioclase and opaque minerals (Fig. 6e).

The granite is pink, and mainly consists of K-feldspar (50–60 vol%), quartz (20–25 vol%) and plagioclase (10–15 vol%); amphibole and biotite are found in some samples. The simultaneous growth of quartz and alkali feldspar in some samples creates a granophyric texture (Fig. 6f). K-feldspar is characterized by platy or anhedral grain-shapes and locally contains euhedral inclusions of plagioclase and quartz. Quartz is subhedral to anhedral and is mainly 0.5–2 mm in size, located between the K-feldspars and

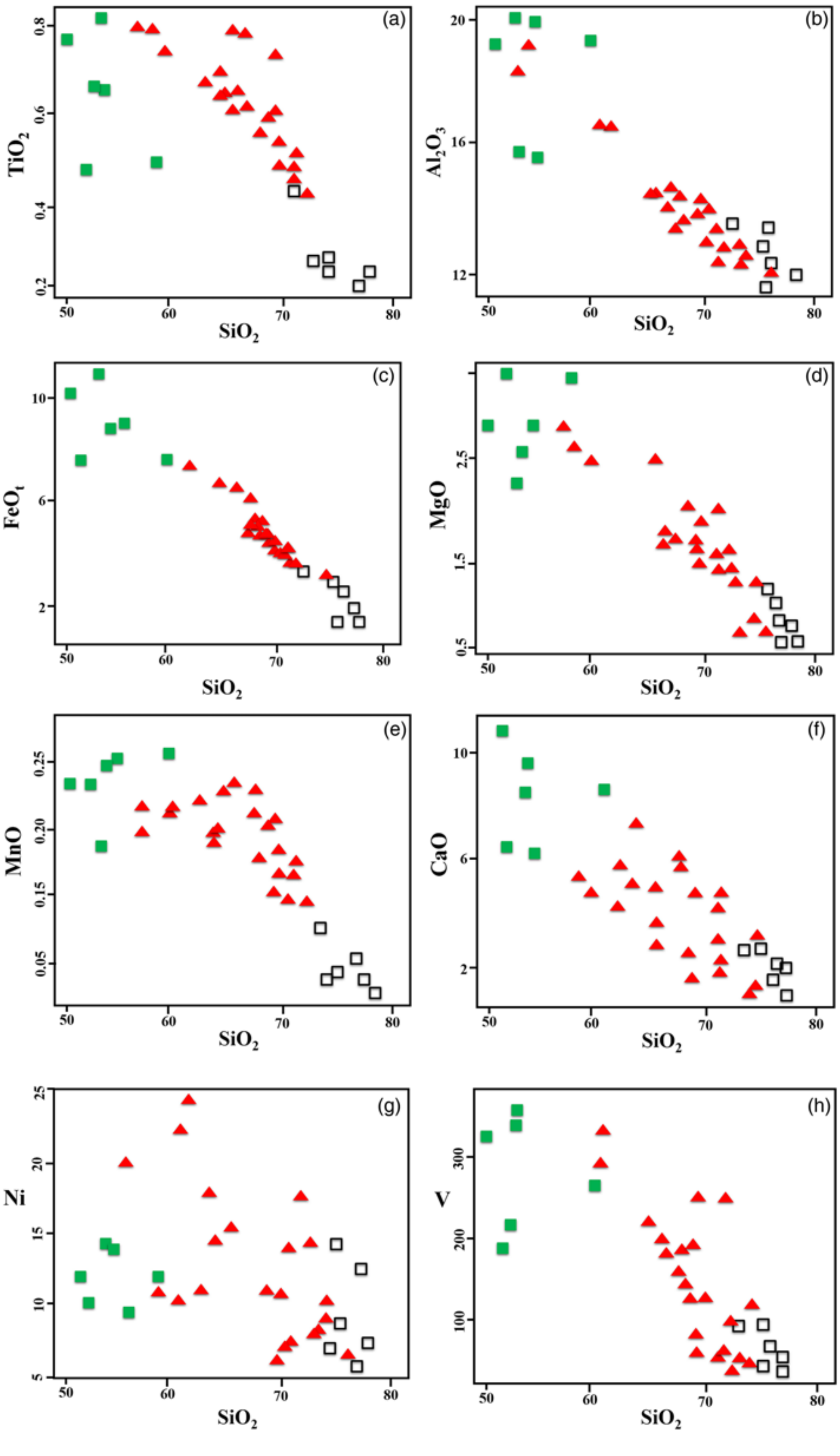

Fig. 9. SiO_2 (wt%) v. major oxides (wt%) and representative trace elements (ppm) for igneous samples from the SW Sabzevar area.

plagioclase. Plagioclase is medium- to fine-grained and shows polysynthetic twinning. In some samples, plagioclase phenocrysts are weakly to moderately altered to sericite, epidote, calcite and chlorite.

Mineral chemistry

Plagioclase is the most abundant mineral in the Late Cretaceous extrusive and intrusive rocks of the SW Sabzevar area. In rhyolites and trachyandesites its composition is mostly albite (Fig. 7a), in the range of $An_{0.54–4.15}$ and $An_{1.96–3.31}$ respectively. In dacites the plagioclase is mostly bytownite in composition, ranging from $An_{74.11}$ in the cores to $An_{63.67}$ in the rims (Fig. 8a). The plagioclase composition ranges from $An_{88.26}$ in the gabbrodiorites to An_{45} in the granites. In general, clinopyroxene phenocrysts do not show compositional zoning. The clinopyroxenes of dacitic samples are augite (Fig. 7b), with a compositional range of $Wo_{43–45}$, $En_{41–42}$, $Fs_{14–17}$. Augite ($Wo_{38–47}$, $En_{42–50}$, $Fs_{6–1}$) is also the dominant clinopyroxene in the trachyandesites (Morimoto *et al.* 1988; Fig. 8b). Clinopyroxene is augite in gabbrodiorites, with a

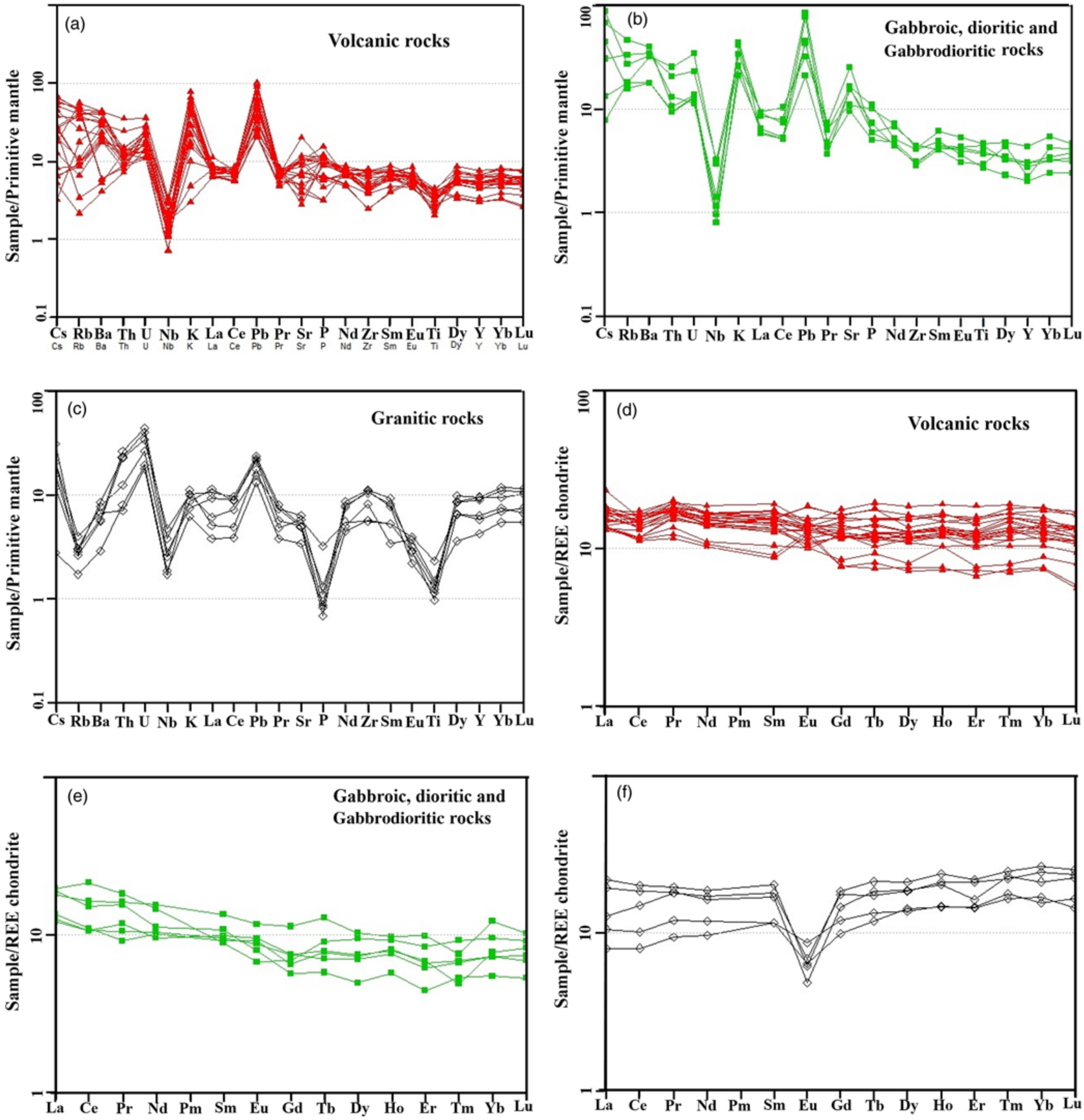

Fig. 10. Spidergrams and REE patterns for the Late Cretaceous volcanic and intrusive rocks from the SW Sabzevar area normalized to the standard values of (**a–c**) primitive mantle (Sun & McDonough 1989), (**d–f**) chondrite (Nakamura 1974).

composition of $Wo_{38–47}$, $En_{42–50}$, $Fs_{6–1}$ (Fig. 7b). The amphiboles from diorites are of igneous origin; their Si values that do not exceed the 7.50 a.p.f.u. (atoms per formula unit) that is the limit of igneous amphiboles (Leake *et al.* 1997). The analytical results were plotted on a diagram from Leake *et al.* (1997) to determine the amphibole type. The amphiboles of dioritic and granitic samples are located in the magnesio-hornblende and tchermakite field, respectively (Fig. 7c). There are no major chemical differences between the cores and rims of the amphiboles.

Whole-rock geochemistry

The intrusive rocks range in composition from gabbro to granite, with SiO_2 content ranging between 50.5 and 78.0 wt%. The extrusive samples range from trachyandesite to rhyolites with SiO_2 contents between 47.7 and 74.9 wt%. Based on the geochemical classification of Le Bas *et al.* (1986), the extrusive samples have trachyandesite, dacite and rhyolite compositions and plot within the sub-alkaline–tholeiitic field (Fig. 8a). The intrusive samples are diorite, gabbrodiorite and granite, in terms of the total alkalis v. SiO_2 contents, and plot in the gabbro, gabbrodiorite and granite fields (Middlemost 1985, Fig. 8b). In the Nb/Y–Ti/Y diagram (Pearce 1982), all of the samples fall in the tholeiitic field (Fig. 8c). Regarding the SiO_2 variation in our samples (47.7–78.0 wt%), the variations of the major elements have been evaluated against SiO_2 (Fig. 9a–h). In the Harker diagrams, the major elements display a good correlation with SiO_2. Major elements TiO_2, Al_2O_3, FeO_t, MgO, CaO and MnO show significant negative correlations with SiO_2 content in all the samples. Ni and V contents also decrease with increasing SiO_2.

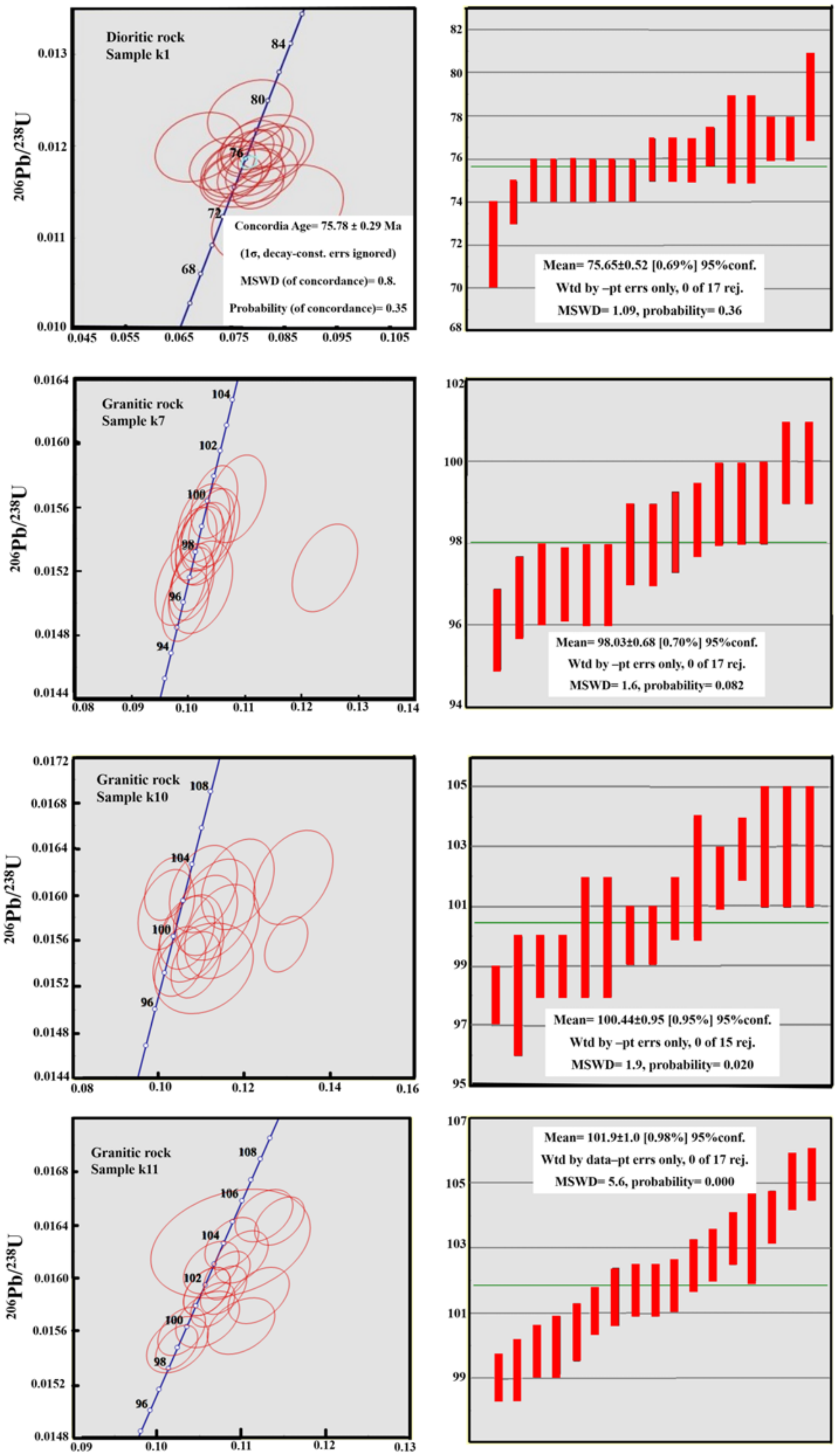

Fig. 11. Concordia diagrams showing the zircon data for the Late Cretaceous intrusive and extrusive rocks from the SW Sabzevar area.

The Sabzevar intrusive and extrusive rocks have different ages and isotopic compositions (see below), and therefore there is no genetic link between them. Therefore, these trends do not show the liquid line of descent among the samples, but instead the plots can be used to discriminate between fairly primitive and fractionated samples.

The chondrite-normalized REE patterns (Fig. 10d–f) for the volcanic rocks are similar and flat, whereas those of mafic to intermediate intrusive rocks show mild enrichment in the light REE (LREE). The REE patterns of the granitic samples show a mild LREE depletion relative to the gabbroic and dioritic samples and have pronounced negative Eu anomalies that are absent in the more mafic intrusive rocks, and all of the volcanic rocks. Primitive-mantle-normalized multi-element diagrams (Fig. 10a–c) show negative Nb and Ti anomalies and positive anomalies in Pb and K. These geochemical signatures indicate that these rocks are calc-alkaline in nature, and their melts were generated from the partial melting of a mantle metasomatized by fluids from a subduction zone.

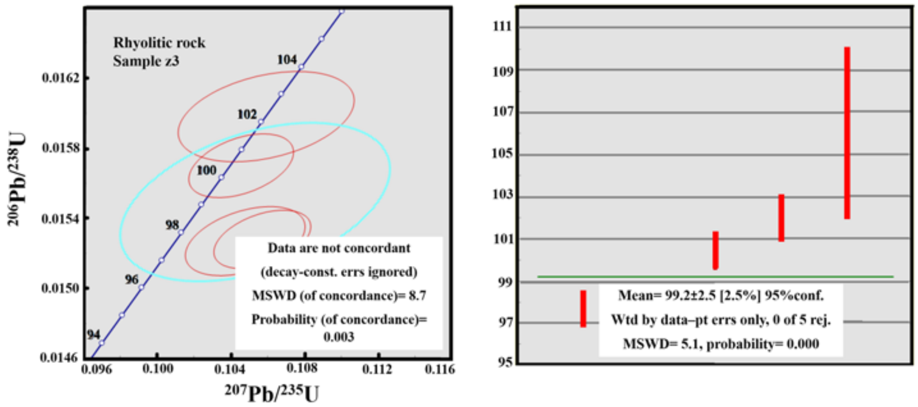

Fig. 11.Continued.

Zircon U–Pb age dating

Three granites (K-7, K-10 and K-11), one diorite (K-2) and one rhyolite (Z-3) were selected for zircon U–Pb age dating (Fig. 11). Zircon grains are mostly subhedral and grains show zoning. Fifteen, 26 and 17 spots analysed on zircon grains from Ka-7, Ka-10 and Ka-11 samples, yielded weighted mean ages of 98.03 ± 0.68 Ma (MSWD = 1.6), 101.9 ± 1 Ma (MSWD = 5.6) and 100.44 ± 0.95 Ma (MSWD = 1.9), respectively. Twenty-two analyses on 22 zircon grains from the dioritic (K-2) sample yielded a concordia age of 75.78 ± 0.29 Ma with MSWD = 0.78. Ten spots were analysed on zircon grains from the rhyolite Z-3; four of these were ignored in age calculations owing to their high discordance, and the remaining six analyses yielded a weighted mean age of 99.2 ± 2.5 Ma with MSWD = 5.1. As shown in Figure 11, several of the ages have large MSWD values, reflecting a scatter of the zircons along the concordia; this may reflect either thermal disturbance or a degree of inheritance and the mixing of magma batches, especially in the most felsic rocks. These ages show that magmatism in NE Iran started at *c.* 102 Ma and lasted for *c.* 26 myr. They show that the Mesozoic crust of NE Iran was exclusively built by a pulsed magmatism for a fairly short time (*c.* 26 myr), which is usual in continental arcs such as the Andes (e.g. DeCelles *et al.* 2009).

Isotopic geochemistry

Initial isotopic values of Sr and Nd in the rhyolite, diorite and three granite samples were calculated using their zircon U–Pb ages (i.e. *c.* 99, 76, 100 and 102 Ma). The results are presented in Table 8. The extrusive rocks are characterized by $(^{87}Sr/^{86}Sr)_i$ ratios from 0.70411 to 0.70628, $(^{143}Nd/^{144}Nd)_i$ values from 0.51282 to 0.51289 and

Table 5. *Zircon U–Pb age dating results for diorite sample K-2*

Analysis	$^{207}Pb/^{206}Pb$	±2σ	$^{207}Pb/^{235}U$	±2σ	$^{206}Pb/^{238}U$	±2σ	$^{208}Pb/^{232}Th$	±2σ
K-2-01	115	130	99	4	97.8	2	92	4
K-2-02	500	60	529	10	536	6	479	18
K-2-03	552	58	542	10	540	6	506	20
K-2-04	166	348	79	12	77	4	71	8
K-2-05	95	162	77	6	76.6	18	75	4
K-2-06	2	312	77	10	79	4	80	8
K-2-07	100	222	77	8	76	2	74	4
K-2-08	591	64	544	12	532	8	490	20
K-2-09	4397	5086	704	500	84	60	65	114
K-2-10	2694	2764	657	534	226	56	198	120.46
K-2-11	−19	250	72	8	76	2	76	6
K-2-12	4865	192	1888	76	321	32	816	62
K-2-13	35	240	12.5	14	12.4	0.4	12.9	1.2
K-2-14	41	264	74	8	75	2	79	8
K-2-15	2691	3392	347	362	102	46	89	60
K-2-16	4155	1044	1017	306	169	58	135	62
K-2-17	546	74	532	14	528	8	526	30
K-2-18	−196	308	68	10	77	4	75	10
K-2-19	3794	1074	929	306	186	56	152	52
K-2-20	3415	2570	598	396	128	56	107	104
K-2-21	46	252	75	8	75	2	71	6
K-2-22	152	216	78	6	75	2	77	6
K-2-23	207	248	81	8	76	2	74	6
K-2-24	207	194	78	6	74	2	75	4
K-2-25	371	470	82	16	72	4	73	10
K-2-26	93	106	96	4	96.5	16	98	4
K-2-27	85	296	78	10	75	2	80	6
K-2-28	39	226	74	8	75	2	74	4
K-2-29	85	232	77	8	77	2	78	6
K-2-30	159	198	80	6	77	2	76	6
K-2-31	101	320	75	10	75	2	76	6

Table 6. *Zircon U–Pb age dating results for granite samples K-11 and K-10*

Analysis	$^{207}Pb/^{206}Pb$	±2σ	$^{207}Pb/^{235}U$	±2σ	$^{206}Pb/^{238}U$	±2σ	$^{208}Pb/^{232}Th$	±2σ
K-11-01	68	142	96	6	97	2	95	8
K-11-02	83	126	98	4	98.3	2	99	6
K-11-03	126	130	101	6	100	2	105	6
K-11-04	209	156	105	6	100	2	98	6
K-11-05	108	120	99	4	98.6	1.8	97	4
K-11-06	108	134	96	6	95.9	2	96	6
K-11-07	102	144	99	6	99	2	104	6
K-11-08	109	120	97	4	97	1.8	95	4
K-11-09	110	138	99	6	98	2	101	6
K-11-10	573	144	119	6	97	2	90	6
K-11-11	152	178	99	6	97	2	101	8
K-11-12	148	142	101	6	99	2	95	6
K-11-13	123	154	99	6	98	2	100	6
K-11-14	125	148	100	6	99	2	95	6
K-11-15	112	130	97	4	96.7	2	99	6
K-10-01	162	192	101	8	99	2	98	6
K-10-02	190	226	104	10	100	4	96	8
K-10-03	4	150	99	6	103	2	96	6
K-10-04	77	382	107	16	108	6	83	12
K-10-05	250	162	106	6	99	2	95	6
K-10-06	332	244	110	10	100	4	107	10
K-10-07	251	174	105	8	98	2	107	8
K-10-08	620	120	124	6	100	2	16	4
K-10-09	234	218	107	10	101	2	107	10
K-10-10	569	220	125	12	103	4	103	10
K-10-11	102	218	106	10	107	4	110	10
K-10-12	346	130	114	10	103	4	108	10
K-10-13	190	230	106	10	103	4	111	10
K-10-14	164	196	103	8	100	2	100	8
K-10-15	294	328	106	14	98	4	97	4
K-10-16	276	264	109	12	102	4	102	10
K-10-17	25	192	98	8	102	2	106	10

Table 7. *Zircon U–Pb age dating results for granite sample K-7 and rhyolite sample Z-3*

Analysis	$^{207}Pb/^{206}Pb$	±2σ	$^{207}Pb/^{235}U$	±2σ	$^{206}Pb/^{238}U$	±2σ	$^{208}Pb/^{232}Th$	±2σ
K-7-01	73	124	105	6	107	2	104	6
K-7-02	123	270	105	12	104	4	104	8
K-7-03	219	84	110	4	105.1	1.6	99	4
K-7-04	147	118	102	4	100.4	1.8	106	4
K-7-05	91	92	99	4	99.8	1.6	97	4
K-7-06	239	118	106	4	100	1.8	111	6
K-7-07	206	78	102	4	97.4	1.4	98	4
K-7-08	99	98	98	4	98	1.6	95	4
K-7-09	114	86	102	4	101.7	1.6	100	4
K-7-10	95	116	99	4	99.2	1.8	95	4
K-7-11	250	104	108	4	101.5	1.8	100	4
K-7-12	96	80	99	4	99	1.4	98	4
K-7-13	144	80	103	4	101.1	1.4	101	4
K-7-14	169	200	101	8	98	2	98	2
K-7-15	126	98	105	4	104	1.6	107	4
K-7-16	165	94	106	4	108.3	1.6	102	4
K-7-17	149	84	104	4	102.5	1.6	104	4
K-7-18	114	86	102	4	101.7	1.6	104	4
K-7-19	98	102	103	4	102.8	1.6	103	6
K-7-20	176	100	108	4	105.3	1.6	107	6
K-7-21	156	84	104	4	101.8	1.6	104	4
Z-3-01	3933	940	927	280	169	48	137	74
Z-3-02	266	286	105	12	98	4	85	8
Z-3-03	3824	1298	822	326	152	50	124	82
Z-3-04	114	90	101	4	100.4	1.6	97	4
Z-3-05	102	144	102	6	102	2	95	4
Z-3-06	203	82	102	4	97.8	1.4	86	2
Z-3-07	184	102	101	4	97.7	1.6	101	4
Z-3-08	157	90	93	4	90.3	1.4	88	4
Z-3-09	2375	606	305	84	106	8	94	26
Z-3-10	3112	644	470	132	114	14	97	34

Table 8. *Results of the isotopic analyses*

No.	Rock type	Sr (ppm)	Rb (ppm)	$^{87}Rb/^{86}Sr$	Error (2σ)	$^{87}Sr/^{86}Sr$	Error (2σ)	$(^{87}Sr/^{86}Sr)_i$	Sm (ppm)	Nd (ppm)	$^{147}Sm/^{144}Nd$	Error (2σ)	$^{143}Nd/^{144}Nd$	Error (2σ)	$(^{143}Nd/^{144}Nd)_i$	εNd
Z-2	Dacite	426	6.89	0.0469	0.023	0.7061	0.000014	0.7063	2.89	10	0.1749	0.006	0.5129	0.000012	0.5128	6.23
Z-4	Rhyolite	98	11	0.3287	0.034	0.7049	0.000012	0.7048	3.22	10.7	0.1815	0.003	0.5130	0.000024	0.5129	7.38
Z-5	Rhyolite	142	5	0.1119	0.001	0.7044	0.000021	0.70445	3.07	10.7	0.1724	0.006	0.5129	0.000017	0.5128	6.34
Z-8	Dacite	134	22	0.4759	0.077	0.7049	0.000010	0.7049	2.84	9.7	0.1768	0.005	0.5129	0.000015	0.5128	5.94
Z-10	Rhyolite	218	4	0.0531	0.009	0.7041	0.000017	0.7041	2.55	8.92	0.1708	0.006	0.5129	0.000023	0.5128	5.91
Z-15	Trachyandesite	69	1.4	0.0587	0.023	0.7047	0.000022	0.7047	3.4	9.7	0.2110	0.006	0.5130	0.000016	05129	6.93
Z-17	Dacite	217	6	0.0801	0.003	0.6905	0.000020	0.7049	2.73	8.9	0.1855	0.002	0.5130	0.000018	0.5128	6.33
Z-21	Dacite	109	16	0.4401	0.002	0.7050	0.000011	0.7050	2.11	6.9	0.1841	0.005	0.5129	0.000020	0.5128	5.95
k-1	Gabbrodiorite	233	21	0.2651	0.021	0.7044	0.000027	0.7042	2.73	9.8	0.1682	0.007	0.5129	0.000010	0.5129	6.24
k-2	Diorite	541	11	0.0601	0.031	0.7042	0.000018	0.7046	1.80	9.2	0.1181	0.006	0.5130	0.000025	0.5129	7.22
k-3	Gabbrodiorite	204	12	0.1642	0.035	0.7046	0.000016	0.7045	1.94	6.6	0.1783	0.003	0.5129	0.000011	0.5128	5.81
k-7	Granite	122	3	0.0529	0.043	0.7053	0.000023	0.7058	2.35	7.5	0.1889	0.007	0.5130	0.000014	0.5129	7.01
k-10	Granite	105	2	0.0525	0.036	0.7053	0.000015	0.7055	4.13	11.8	0.2113	0.004	0.5130	0.000026	0.5128	6.13
k-11	Granite	135	1.8	0.0386	0.023	0.7059	0.000013	0.7053	3.66	10.8	0.2040	0.005	0.5130	0.000013	0.5128	6.16

εNd values from 5.91 to 7.38. The intrusive rocks have $(^{87}Sr/^{86}Sr)_i$ ratios from 0.70423 to 0.70579, $(^{143}Nd/^{144}Nd)_i$ values from 0.51283 to 0.51292 and εNd values from 5.81 to 7.22. The similarity of the isotopic ratios for the coeval granites and rhyolites is consistent with a genetic relationship, whereas the diorites are younger (*c.* 75 Ma) and probably have no genetic link with the granites and rhyolites. However, the positive εNd and the low and slightly variable $(^{87}Sr/^{86}Sr)_i$ ratios suggest a juvenile and relatively homogeneous source for the volcanic and intrusive rocks. The similarities in the isotopic composition of granites–rhyolites and diorites indicate a similar source for these rocks.

Discussion

Genesis of the igneous rocks

The correlation of decreasing TiO_2, Al_2O_3, Fe_2O_3, MgO, CaO, MnO, Ni and V concentrations with increasing SiO_2 contents in granites and rhyolites is consistent with fractional crystallization (Fig. 9d–f) of plagioclase, amphibole, clinopyroxene and accessory apatite in the samples. However, the diorites are about *c.* 25 myr younger than rhyolites and granites and therefore there is no direct genetic linkage between them. The felsic rocks (granites and rhyolites) can be generated from fractional crystallization of a mafic melt but not from the dioritic melts.

Trace-element patterns of the Late Cretaceous intrusive and extrusive rocks from the SW Sabzevar area show remarkable geochemical consistency, implying that their parental magmas formed by partial melting of a relatively homogeneous mantle source, and that the felsic rocks may be related to the mafic rocks through simple fractionation processes. A negative Eu anomaly is observed in the granitic rocks, but not in other samples. The removal of a small amount of plagioclase through fractional crystallization, or partial melting that leaves plagioclase in the source, can produce such anomalies in the residual melt (Azer 2007). Because of the genetic association of gabbrodiorites with granites apparent in the field, it could be argued that the negative Eu anomaly in the granitic rocks arose from plagioclase fractionation from the parental magma. Unlike mid-ocean ridge basalts (MORB) and ocean island basalts (OIB), the magmas produced in subduction zones are especially rich in large ion lithophile elements (LILE) and LREE and depleted in high field strength elements (HFSE), such as Ta, Zr, Ti and Nb, and heavy rare earth elements (HREE), such as Lu, Yb and transition elements (Woodhead *et al.* 1993). As shown in Figure 10, all samples show some characteristics of subduction- or arc-zone magmas, especially the negative anomalies in Nb and Ti. These anomalies are classically considered to be the result of the release of fluids from the oceanic slab into the mantle wedge during the generation of magmas (e.g. Pearce 1982). However, for highly evolved granites, the removal of HFSE can be also related to the extreme fractionation of accessory minerals such as ilmenite and titanite.

HFSE such as Zr, Y, Nb, Ti and P, and middle REE (MREE), HREE and the transition metals are immobile during secondary alteration. Therefore, the geochemical characteristics and tectonic setting of the analysed samples can be elucidated on the basis of these immobile elements. According to Dai *et al.* (2011), magmas derived from the mantle have a low ratio of Lu/Yb with an average of 0.14–0.15, whereas this ratio is higher in the continental crust (0.16–0.18). The average ratio of Lu/Yb in all studied samples is *c.* 0.14, which implies that these rocks have no evidence of contamination with continental crust. The subduction signature of the studied extrusive and intrusive rocks is confirmed by the Ta–Yb, Ta + Yb–Rb (Pearce *et al.* 1984), Ta/Yb–Th/Yb (Pearce 2008), Th/Yb–La/Yb (Pearce 1982) and Th/Yb v. Ta/Yb (Brown *et al.* 1984) diagrams, in which all of the samples plot in the arc-related

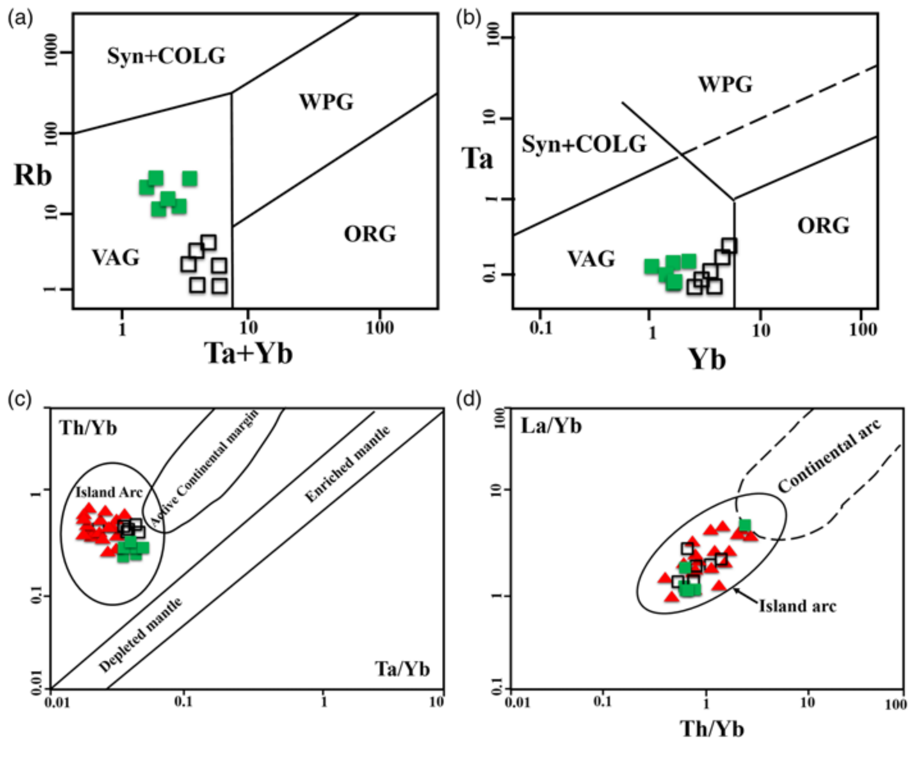

Fig. 12. The position of the Late Cretaceous igneous rocks of the SW Sabzevar area in various tectonomagmatic diagrams: (**a**) and (**b**) Ta + Yb–Rb and Ta–Rb (Pearce *et al.* 1984) (ORG, ocean ridge granites; Syn-COLG, syn-collision granites; VAG, volcanic arc granites; WPG, within-plate granites); (**c**) Ta/Yb–Th/Yb (Pearce 2008); (**d**) Th/Yb–La/Yb (Pearce 1982) for the Late Cretaceous igneous rocks of SW Sabzevar area.

field (Fig. 12a–d). These data clearly imply that the Late Cretaceous igneous rocks from the SW Sabzevar region display a subduction signature.

The trace elements and their ratios also can be used to determine the contributions of the subducting slab and the mantle in the genesis of the magmas (Hawkesworth *et al.* 1993). In particular, the ratio of the mobile to immobile incompatible elements can be used to assess the influence of the slab-derived fluids on arc magmas. The ratios of trace elements to SiO_2 can be used to assess the effects of fractional crystallization. Negative correlation of some selected ratios of trace elements, such as U/Nb, Pb/Nb and Ba/Nb, with SiO_2 may show that continental-crust contamination played a major role during the genesis of a series of rocks. These trace element ratios show little or no correlation with SiO_2 content for the Sabzevar non-ophiolitic rocks (Fig. 13a–c), which suggests no conspicuous crustal contamination, but instead indicates that the source(s) of these rocks has been modified by fluids released from the subducted slab (Plank 2005).

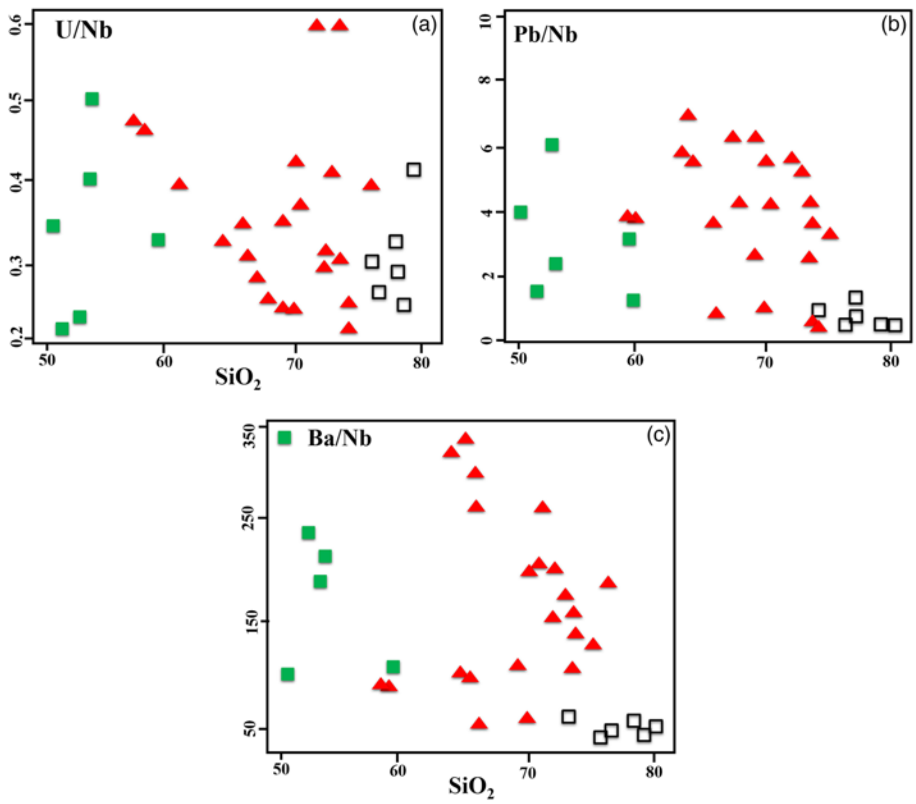

Fig. 13. (**a**–**c**) Plots of the Late Cretaceous igneous rocks of the SW Sabzevar area in plots of fluid-mobile/immobile incompatible trace elements ratio (U/Nb, Pb/Nb andBa/Nb) v. SiO_2 contents to demonstrate the limited effect ofmagma differentiation on these ratios.

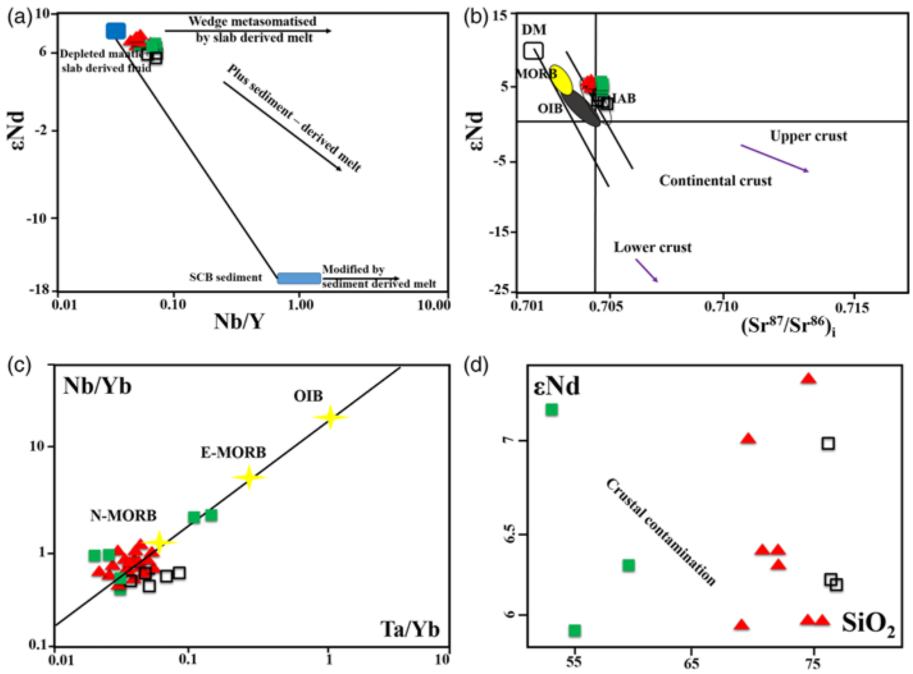

Fig. 14. (**a**) The εNdv. Nd/Y (Wang *et al.* 2013) plot for the Late Cretaceous igneous rocks of the SW Sabzevar area. The compositions of the end-members are from Elliott *et al.* (1997), Class *et al.* (2000), Castillo *et al.* (2007), Petrone & Ferrari (2008) and Wang *et al.* (2013). SCB; South China Block. (**b**) εNd–($^{87}Sr/^{86}Sr$)$_i$ diagram. The analysed samples plot in the depleted zone. (**c**) Ta/Yb v. Nb/Yb diagram (Pearce 1982; Pang *et al.* 2013). In this diagram all samples plot near N-MORB. (**d**) SiO_2 v. εNd (Xu *et al.* 2014) for the analysed samples.

In the Nb/Y–εNd diagram, all of the studied samples show low Nb/Y ratios (Wang *et al.* 2013; Fig. 14a), another indicator of the role of released slab-derived fluids in the magma genesis.

As expected, elements enriched in arc magmas are usually found in large amounts in the continental crust. The enrichment and depletion of elements in arc magmas reflect processes in the source, and particularly the addition of subducted components (Hawkesworth *et al.* 1993). Volcanic arcs can be classified into highly enriched and poorly enriched categories based on Ce/Yb ratios (Juteau & Maury 1997); enriched arcs are defined as having Ce/Yb > 15. The mean Ce/Yb of the rocks studied is <5 and defines a poorly enriched arc signature. The placement of samples near the mantle array and positive εNd values indicate that they have originated from the depleted mantle and out of the continental crust (Fig. 14b). All intrusive and extrusive samples have similar Ta/Yb and Nb/Yb ratios and plot near the N-MORB field, indicating that crystal fractionation of amphibole and/or titanite–rutile was negligible during their magmatic evolution (Pearce 1982; Pang *et al.* 2013; Fig. 14c). The effect of crustal contamination is debatable; the εNd v. SiO_2 plot (Xu *et al.* 2014) does not show an obvious crustal-addition trend, except perhaps in the scatter among the felsic samples. The lack of correlation between the isotopic ratios and SiO_2 in the rocks of the study area would imply the absence of significant crustal contamination (Ma *et al.* 2014; Fig. 14d). In summary, these characteristics imply a depleted mantle, enriched by subduction-related fluids without continental components, as the source of the primary magma. However, it should be noted that all the geochemical signatures mentioned above for the Sabzevar igneous rocks, including high εNd, can also indicate that these rocks were generated by partial melting of a juvenile arc crust. This crust could be formed in the early stages of the subduction initiation and nascent arc magmatism. If this crust

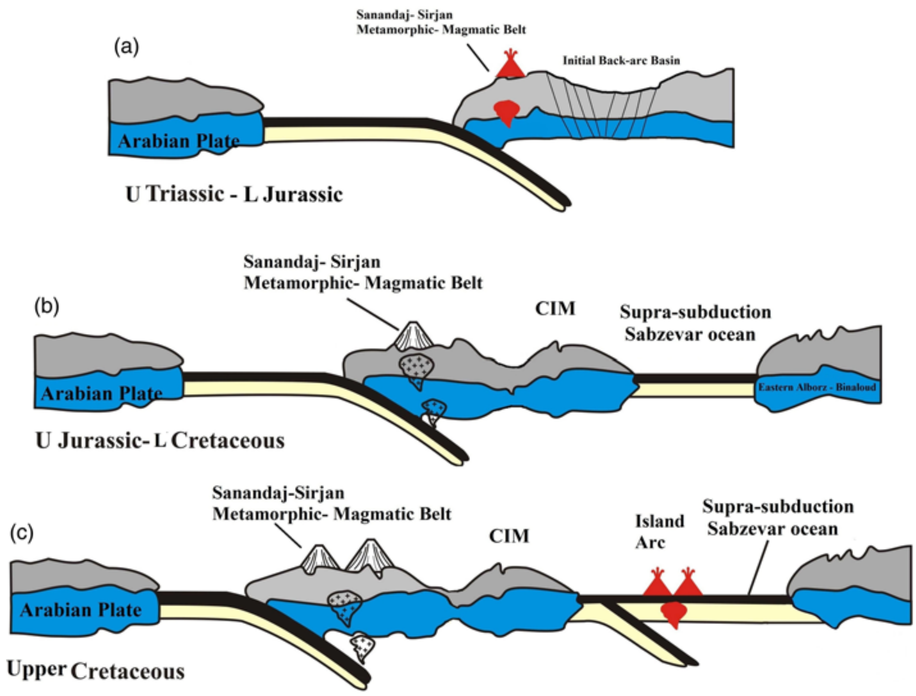

Fig. 15. A schematic tectonic model for development of the Late Cretaceous magmatism of the SW Sabzevar area. A, Late Triassic–Early Jurassic; B, Late Jurassic–Late Cretaceous; C, Late Cretaceous.

melts, it will generate rocks that are geochemically similar to arc rocks and have high εNd isotope values. This can be an alternative model for the formation of the Sabzevar late Cretaceous non-ophiolitic rocks, but we need more age and isotopic studies on the early arc mafic rocks to understand this possibility.

Tectonomagmatic model

Most Iranian ophiolites are remnants of the Neotethys oceanic lithosphere and are thus related to East Mediterranean and North Indian–Himalayan ophiolites. The occurrence of these ophiolites is suggested to be related to the initiation of Neotethyan subduction along the southern margin of Eurasia (e.g. Dilek & Furnes 2009; Pearce & Robinson 2010; Moghadam & Stern 2014). The Neotethyan Ocean opened in the late Paleozoic (Permian) to middle Triassic between Gondwana and the Cimmerian continental blocks that today make up Anatolia, Iran and Afghanistan, which were separated from Gondwana (Dercourt *et al.* 1986). Most researchers believe that the subduction of the Neotethyan Ocean began in the middle–Late Triassic (Berberian & King 1981; Agard *et al.* 2005; Bagheri & Stampfli 2008; Omrani *et al.* 2008; Chiu *et al.* 2013; Mehdipour & Moazzen 2015), although there is no conspicuous Triassic magmatism in Iran to support this idea. There are also different ideas about the Neotethyan closure time, which include late Cretaceous (Berberian & King 1981), Late Paleocene–early Eocene (Mazhari *et al.* 2009), late Eocene–Oligocene (Dargahi *et al.* 2010; Agard *et al.* 2011) and even middle Miocene–Pliocene (McQuarrie *et al.* 2003; Azizi & Moine Vaziri 2009). Late Cretaceous is suggested to be the time of ophiolite obduction along the passive margin of Arabia (e.g. Ricou 1968; Golonka 2004).

A Late Eocene–Oligocene (30 ± 5 Ma) timing, which is consistent with extension, core-complex formation and rapid exhumation of the Iranian plateau, better explains the time of Neotethyan closure (McQuarrie & van Hinsbergen, 2013). Moreover, the occurrence of *c.* 11 Ma post-collisional ultrapotassic rocks and vestiges of *c.* 25 Ma diatexites in NW Iran suggest that collision was under way in Late Oligocene–Early Miocene time (Moghadam *et al.* 2014, 2016). Therefore, the final Neotethyan closure seems to be diachronous along the Main Zagros Thrust, implying that the collision in Iran began at *c.* 30 Ma and that collisional shortening was at a maximum by *c.* 20 Ma.

The subduction of Neotethys was suggested to occur in Jurassic to early Cretaceous time, along with the formation of a cordilleran margin along the Sanandaj–Sirjan zone (Berberian & King 1981; Ghasemi & Talbot 2006; Omrani *et al.* 2008). However, the Sanandaj–Sirjan zone is also interpreted as a continental rift, as igneous rocks from this zone show rift-related geochemical signatures and magmatism younging from SE to NW, which is broadly consistent with plume-related rift activity (e.g. Hunziker *et al.* 2015; Azizi *et al.* 2018). The Late Cretaceous is considered to mark the initiation of subduction of the Neotethyan ocean beneath Iran (Eurasia), which is marked by Zagros suprasubduction-zone-related ophiolites (Moghadam & Stern 2014). The initiation of subduction was accompanied by hyperextension in the Iranian plateau, which formed the Late Cretaceous back-arc basins including the Sabzevar Ocean in the NE (Sengor *et al.* 1988; Richards & Sholeh 2016). These oceanic basins began to open in the early late Cretaceous (*c.* 104 Ma), expanded significantly during the Late Cretaceous and were closing by latest Cretaceous–Paleocene time (Baroz *et al.* 1983; Shojaat *et al.* 2003). Numerous researchers have examined the magmatic belt in NE Iran, along the Sabzevar zone. The northward subduction of the Sabzevar oceanic crust, beneath the Turan block, has been mentioned in various studies. The Sabzevar ophiolite complex, as a remnant of this ocean, was formed during the initiation of northward, intra-oceanic subduction of the Sabzevar Ocean in the late Cretaceous and the closure of this waterway in the latest Cretaceous–Early Paleocene time. According to Maghfouri *et al.* (2016), the submarine magmatism (intense calc-alkaline to tholeiitic) of the SW Sabzevar area was associated with the northward subduction of the Sabzevar back-arc basin crust beneath the southern margin of the Turan block.

Zircon U–Pb age dating of granitoids and volcanic rocks from the SW Sabzevar area has given Late Cretaceous ages (102–76 Ma), when the Sabzevar oceanic crust was subducting and new island arcs were forming. The collision of Central Iran with the Turan block during the Paleogene resulted in closure of the Sabzevar Ocean and obduction of ophiolites around the Central Iran microcontinent. The magmatism of the area as studied here is island-arc tholeiitic to calc-alkaline in the south; it migrated toward the north and NE, evolving toward a calc-alkaline continental margin arc with time. The adakitic nature of Eocene magmatism in the Sabzevar ophiolite belt (Ghasemi & Rezaei 2015; Jamshidi *et al.* 2015) and Miocene adakitic magmatism in Quchan (north of the Sabzevar ophiolites) (Shabanian *et al.* 2012) indicates the incidence of magmatism in a continental margin arc, although Neogene adakites may indicate a slab break-off rather than magmatism in a continental margin arc.

The opening of the Sabzevar Ocean basin from the early late Cretaceous and its expansion during the Late Cretaceous have been associated with the tholeiitic basaltic magmatism of a mid-ocean ridge (N-MORB). Also, the beginning of subduction in the latest Cretaceous has been associated with the island-arc tholeiitic and calc-alkaline magmatism (Fig. 15).

Conclusions

The Late Cretaceous volcanic and plutonic igneous rocks of the SW Sabzevar area include an assemblage of trachyandesites, dacites, rhyolites, gabbros, gabbrodiorites, diorites and granites. These rocks, which occur within the Late Cretaceous sedimentary rocks, especially in *Globotruncana*-bearing limestones and pyroclastic rocks, have island-arc tholeiitic and calc-alkaline signatures. The tholeiitic parental magma has been generated by partial melting of a depleted source (MORB-type slab or depleted-mantle source), which was metasomatized by fluids derived from dehydration of the Neotethyan oceanic slab. During the Late Cretaceous–Eocene, the ultimate closure of the Sabzevar oceanic basin resulted in the eruption of high-K calc-alkaline rocks and adakites that were followed by the Quaternary adakites in the northern part of the Sabzevar zone, south of Quchan.

Acknowledgements This paper is the result of the PhD project of Z.K. Special thanks go to the staff of the CCFS/GEMOC for kind support during her stay. We thank M. Bebbington for her help with thin-section preparation. This is contribution 1199 from the ARC Centre of Excellence for Core to Crust Fluid Systems (www.ccfs.mq.edu.au) and 1249 from the GEMOC Key Centre (www.gemoc.mq.edu.au).

Funding Z.K. would like to thank the Ministry of Sciences, Research and Technology of Iran and Shahrood University of Technology for the financial support for a visiting scholar period in Macquarie University, Sydney, Australia. The authors are particularly grateful to the ARC Centre of Excellence for Core to Crust Fluid Systems (CCFS) and GEMOC ARC National Key Centre (GEMOC), Macquarie University for its funding of instrument time. This study used instrumentation funded by ARC LIEF and DEST Systemic Infrastructure Grants, Macquarie University and industry.

Scientific editing by Karel Schulmann

References

Agard, P., Omrani, J., Jolivet, L. & Mouthereau, F. 2005. Convergence history across Zagros (Iran): Constraints from collisional and earlier deformation. *International Journal of Earth Sciences*, **94**, 401–419, https://doi.org/10.1007/s00531-005-0481-4

Agard, P., Omrani, J. *et al.* 2011. Zagros orogeny: a subduction-dominated process. *Geological Magazine*, **148**, 1–34.

Alaminia, Z., Karimpour, M.H., Homam, M. & Finger, F. 2013. The magmatic record in the Arghash region (northeast Iran) and tectonic implications. *International Journal of Earth Sciences*, **102**, 1603–1625, https://doi.org/10.1007/s00531-013-0897-1

Andersen, T. 2002. Correction of common lead in U–Pb analyses that do not report ^{204}Pb. *Chemical Geology*, **192**, 59–79, https://doi.org/10.1016/S0009-2541(02)00195-X

Aulbach, S., Griffin, W.L., O'Reilly, S.Y. & McCandless, T. 2004. Genesis and evolution of the lithospheric mantle beneath the Buffalo Head Terrane, Alberta (Canada). *Lithos*, **77**, 413–451, https://doi.org/10.1016/j.lithos.2004.04.020

Azer, M.K. 2007. Tectonic significance of Late Precambrian calc-alkaline and alkaline magmatism in Saint Katherina area, southern Sinai, Egypt. *Geologica Acta*, **5**, 255–272, https://doi.org/10.1344/105.000000298

Azizi, H. & Moine Vaziri, H. 2009. Review of the tectonic setting of Cretaceous to Quaternary volcanism in northwestern Iran. *Journal of Geodynamics*, **47**, 167–179, https://doi.org/10.1016/j.jog.2008.12.002

Azizi, H., Lucci, F., Stern, R., Hasannejad, S. & Asahara, Y. 2018. The Late Jurassic Panjeh submarine volcano in the northern Sanandaj–Sirjan Zone, northwest Iran: Mantle plume or active margin? *Lithos*, **308–309**, 364–380, https://doi.org/10.1016/j.lithos.2018.03.019

Bagheri, S. & Stampfli, G.M. 2008. The Anarak, Jandaq and Poshte-Badam metamorphic complex in central Iran: New geological data, relationships and tectonic implications. *Tectonophysics*, **451**, 123–155, https://doi.org/10.1016/j.tecto.2007.11.047

Baroz, F., Macaudiere, J., Montigny, R., Noghreyan, M., Ohnenstetter, M. & Rocci, G. 1983. *Ophiolites and related formations in the central part of the Sabzevar range (Iran) and possible geotectonic reconstructions*. Geological Survey of Iran, Report, **51**, 205–237.

Barrier, E. & Vrielynck, B. 2008. *Paleotectonic maps of the Middle East: Atlas of 14 maps*. Middle East Basin Evolution (MEBE) Programme, CGMW, Paris, 1–14.

Bauman, A., Spies, O. & Lensch, G. 1983. *Strontium isotopic composition of post-ophiolitic Tertiary volcanics between Kashmar, Sabzevar and Quchan NE Iran. Geodynamic project (geotraverse) in Iran, Final report*. Geological Survey of Iran, Report, **51**, 342–375.

Berberian, M. & King, G.C.P. 1981. Toward a paleogeography and tectonic evolution of Iran. *Canadian Journal of Earth Sciences*, **18**, 210–265, https://doi.org/10.1139/e81-019

Brown, G.C., Thorpe, R.S. & Webb, P.C. 1984. The geochemical characteristics of granitoids in contrasting arcs and comments on magma sources. *Journal of the Geological Society, London*, **141**, 413–426, https://doi.org/10.1144/gsjgs.141.3.0413

Cao, M., Qin, K., Li, G., Evans, N., Hollings, P., Maisch & Kappler, A. 2017. Mineralogical evidence for crystallization conditions and petrogenesis of ilmenite-series I-type granitoids at the Baogutu reduced porphyry Cu deposit (Western Junggar, NW China): Mössbauer spectroscopy, EPM and LA-(MC)-ICPMS analyses. *Ore Geology Reviews*, **86**, 382–403, https://doi.org/10.1016/j.oregeorev.2017.02.033

Castillo, P.R., Rigby, S.J. & Solidum, R.U. 2007. Origin of high field strength element enrichment in volcanic arcs: geochemical evidence from the southern Sulu Arc, southern Philippines. *Lithos*, **97**, 271–288, https://doi.org/10.1016/j.lithos.2006.12.012

Chang, Z.S., Vervoort, J.D., McClelland, W.C. & Knaack, C. 2006. U–Pb dating of zircon by LA-ICP-MS. *Geochemistry, Geophysics, Geosystems*, **7**, https://doi.org/10.1029/2005GC001100

Chiu, H.Y., Chung, S.L., Zarrinkoub, M.H., Mohammadi, S.S., Khatib, M.M. & Iizuka, Y. 2013. Zircon U–Pb age constraints from Iran on the magmatic evolution related to Neotethyan subduction and Zagros orogeny. *Lithos*, **162–163**, 70–87, https://doi.org/10.1016/j.lithos.2013.01.006

Class, C., Miller, D.M., Goldstein, S.L. & Langmuir, C.H. 2000. Distinguishing melt and fluid subduction components in Umnak Volcanics, Aleutian Arc. *Geochemistry, Geophysics, Geosystems*, **1**, 1004, https://doi.org/10.1029/1999GC000010

Dai, J., Wang, C. *et al.* 2011. Late Devonian OIB alkaline gabbro in the Zangbo Zone: Remnants of the Paleo-Tethys? *Gondwana Research*, **19**, 232–243, https://doi.org/10.1016/j.gr.2010.05.015

Dargahi, S., Arvin, M., Pan, Y. & Babaei, A. 2010. Petrogenesis of post-collisional A-type granitoids from the Urumieh–Dokhtar magmatic assemblage, Southwestern Kerman, Iran: constraints on the Arabian–Eurasian continental collision. *Lithos*, **115**, 190–204, https://doi.org/10.1016/j.lithos.2009.12.002

DeCelles, P., Ducea, M., Kapp, P. & Zandt, G. 2009. Cyclicity in Cordilleran orogenic systems. *Nature Geoscience*, **2**, 251–257, https://doi.org/10.1038/ngeo469

Deer, W.A., Howie, R.A. & Zussman, J. 1966. *An Introduction to the Rock-Forming Minerals*. Longman, Harlow.

Dercourt, J., Zonenshain, L.P. *et al.* 1986. Geological evolution of the Tethys belt from the Atlantic to the Pamir since the Lias. *Tectonophysics*, **123**, 241–315, https://doi.org/10.1016/0040-1951(86)90199-X

Dilek, Y. & Furnes, H. 2009. Structure and geochemistry of Tethyan ophiolites and their petrogenesis in subduction rollback systems. *Lithos*, **113**, 1–20, https://doi.org/10.1016/j.lithos.2009.04.022

Elliott, T., Plank, T., Zindler, A., White, W. & Bourdon, B. 1997. Element transport from slab to volcanic front at the Mariana Arc. *Journal of Geophysical Research*, **102**, 997–1019, https://doi.org/10.1029/97JB00788

Ghasemi, H. & Rezaei, M. 2015. Petrochemistry and tectonic setting of the Davarzan–Abbasabad Eocene Volcanic (DAEV) rocks, NE Iran. *Mineralogy and Petrology*, **109**, 235–252, https://doi.org/10.1007/s00710-014-0353-3

Ghasemi, A. & Talbot, C.J. 2006. A new tectonic scenario for the Sanandaj–Sirjan Zone, Iran. *Journal of Asian Earth Sciences*, **26**, 683–693, https://doi.org/10.1016/j.jseaes.2005.01.003

Golonka, J. 2004. Plate tectonic evolution of the southern margin of Eurasia in the Mesozoic and Cenozoic. *Tectonophysics*, **381**, 235–273, https://doi.org/10.1016/j.tecto.2002.06.004

Griffin, W.L., Powell, W.J., Pearson, N.J. & O'Reilly, S.Y. 2008. GLITTER: data reduction software for laser ablation ICP-MS. *In*: Sylvester, P. (ed.) *Laser Ablation-ICP-MS in the Earth Sciences: Current Practices and Outstanding Issues*. Mineralogical Association of Canada, Short Course Series, **40**, 204–207.

Hawkesworth, C.J., Gallagher, K., Hergt, J.M. & McDermott, F. 1993. Mantle slab contributions in arc magmas. *Annual Review of Earth and Planetary Sciences*, **21**, 175–204, https://doi.org/10.1146/annurev.ea.21.050193.001135

Hosseini, M., Hassanzadeh, J., Alirezaei, S., Sun, W. & Li, Y. 2017. Age revision of the Neotethyan arc migration into the southeast Urumieh–Dokhtar belt of Iran: Geochemistry and U–Pb zircon geochronology. *Lithos*, **284–285**, 296–309, https://doi.org/10.1016/j.lithos.2017.03.012

Hunziker, D.J.P., Burg, P. & Quadt, A. 2015. Jurassic rifting at the Eurasian Tethys margin: Geochemical and geochronological constraints from granitoids of North Makran, southeastern Iran. *Tectonics*, **34**, 571–593, https://doi.org/10.1002/2014TC003768

Jackson, S.E., Pearson, N.J., Griffin, W.L. & Belousova, E.A. 2004. The application of laser ablation-inductively coupled plasma-mass spectrometry to *in situ* U–Pb zircon geochronology. *Chemical Geology*, **211**, 47–69, https://doi.org/10.1016/j.chemgeo.2004.06.017

Jamshidi, K., Ghasemi, H., Troll, V.R., Sadeghian, M. & Dahren, B. 2015. Magma storage and plumbing of adakite-type post-ophiolite intrusions in the Sabzevar ophiolitic zone, northeast Iran. *Solid Earth*, **6**, 49–72, https://doi.org/10.5194/se-6-49-2015

Juteau, T. & Maury, R. 1997. *Géologie de la Croûte océanique: Pétrologie et Dynamique endogène*. Masson, Paris.

Leake, B.E., Woolley, A.R. *et al.* 1997. Nomenclature of amphiboles: Report of the Subcommittee on Amphiboles of the International Mineralogical Association, Commission on New Minerals and Mineral Names. *American Mineralogist*, **82**, 1019–1037.

Le Bas, M.J., Le Maitre, R.W., Streckeisen, A. & Zanettin, B. 1986. A chemical classification of volcanic rocks based on the total alkali–silica diagram. *Journal of Petrology*, **27**, 745–750, https://doi.org/10.1093/petrology/27.3.745

Lindenberg, H.G., Gorler, K. & Ibbeken, H. 1983. *Stratigraphy, structure and orogenetic evolution of the Sabzevar zone, the area of Oryan Khorasan, NE Iran*. Geological Survey of Iran, Report, **51**, 120–142.

Ludwig, K.R. 2003. *User's Manual for Isoplot 3.00: A Geochronological Toolkit for Microsoft Excel*. Berkeley Geochronology Center Special Publication **4**.

Ma, L., Jiang, S. *et al.* 2014. Geochemistry of early Cretaceous calc-alkaline lamprophyres in the Jiaodong Peninsula: Implication for lithospheric evolution of the eastern North China craton. *Gondwana Research*, **25**, 859–872, https://doi.org/10.1016/j.gr.2013.05.012

Maghfouri, S., Rastad, E., Mousivand, F., Lin, Y. & Zaw, K. 2016. Geology, ore facies and sulfur isotopes geochemistry of the Nudeh Besshi-type volcanogenic massive sulfide deposit, southwest Sabzevar basin, Iran. *Journal of Asian Earth Sciences*, **125**, 1–21, https://doi.org/10.1016/j.jseaes.2016.04.022

Mazhari, S.A., Bea, F. *et al.* 2009. The Eocene bimodal Piranshahr massif of the Sanandaj–Sirjan Zone, NW Iran. A marker of the end of the collision in the Zagros Orogen. *Journal of the Geological Society, London*, **166**, 53–69, https://doi.org/10.1144/0016-76492008-022

McPhie, J., Doyle, M. & Allen, R. 1993. *Volcanic Textures: a Guide to the Interpretation of Textures in Volcanic Rocks*. University of Tasmania Centre for Ore Deposit and Exploration Studies, Hobart.

McQuarrie, N. & van Hinsbergen, D. 2013. Retrodeforming the Arabia–Eurasia collision zone: Age of collision versus magnitude of continental subduction. *Geology*, **41**, 315–318, https://doi.org/10.1130/G33591.1

McQuarrie, N., Stock, J.M., Verdel, C. & Wernicke, B.P. 2003. Cenozoic evolution of Neotethys and implications for the causes of plate motions. *Geophysical Research Letters*, **30**, 20–36, https://doi.org/10.1029/2003GL017992

Mehdipour, G.J. & Moazzen, M. 2015. Geodynamic evolution of the Sanandaj–Sirjan zone, Zagros orogen, Iran. *Turkish Journal of Earth Sciences*, **24**, 513–528, https://doi.org/10.3906/yer-1404-12

Middlemost, E.A.K. 1985. *Magmas and Magmatic Rocks*. Longman, Harlow.

Moghadam, H.S. & Stern, R. 2014. Ophiolites of Iran: Keys to understanding the tectonic evolution of SW Asia: (I) Paleozoic ophiolites. *Journal of Asian Earth Science*, **91**, 19–38, https://doi.org/10.1016/j.jseaes.2014.04.008

Moghadam, H.S., Corfu, F., Chiaradia, M., Stern, R.J. & Ghorbani, G. 2014. Sabzevar Ophiolite, NE Iran: Progress from embryonic oceanic lithosphere into magmatic arc constrained by new isotopic and geochemical data. *Lithos*, **210–211**, 224–241, https://doi.org/10.1016/j.lithos.2014.10.004

Moghadam, H.S., Kheder, M., Arai, S., Stern, R., Ghorbani, G., Tamura, A. & Ottley, C.H. 2015. Arc-related harzburgite–dunite–chromitite complexes in the mantle section of the Sabzevar ophiolite, Iran: A model for formation of podiform chromitites. *Gondwana Research*, **27**, 575–593, https://doi.org/10.1016/j.gr.2013.09.007

Moghadam, H.S., Rossetti, F. *et al.* 2016. The calc-alkaline and adakitic volcanism of the Sabzevar structural zone (NE Iran): implications for the Eocene magmatic flare-up in Central Iran. *Lithos*, **248**, 517–535, https://doi.org/10.1016/j.lithos.2016.01.019

Moghadam, H.S., Griffin, W. *et al.* 2017. Crustal evolution of NW Iran: Cadomian arcs, Archean fragments and the Cenozoic magmatic flare-up. *Journal of Petrology*, **58**, 2143–2190, https://doi.org/10.1093/petrology/egy005

Morimoto, N., Fabries, J. *et al.* 1988. Nomenclature of pyroxenes. *American Mineralogist*, **62**, 53–62.

Nakamura, N. 1974. Determination of REE, Ba, Fe, Mg, Na and K in carbonaceous and ordinary chondrites. *Geochimica et Cosmochimica Acta*, **38**, 757–775, https://doi.org/10.1016/0016-7037(74)90149-5

Nelson, S.T. & Montana, A. 1992. Sieve-textured plagioclase in volcanic rocks produced by rapid decompression. *American Mineralogist*, **77**, 1242–1249.

Neumann, E., Griffin, W., Pearson, N. & O'Reilly, S. 2004. The evolution of the upper mantle beneath the Canary Islands: information from trace elements and Sr isotope ratios in minerals in mantle xenoliths. *Journal of Petrology*, **45**, 2573–2612, https://doi.org/10.1093/petrology/egh063

Norman, M.D., McCulloch, M.T. & O'Neill, H.S.C. 2006. Magnesium isotopic analysis of olivine by laser-ablation multi-collector ICP-MS: composition dependent matrix effects and a comparison of the Earth and Moon. *Journal of Analytical Atomic Spectrometry*, **21**, 50–54, https://doi.org/10.1039/B510719A

Omrani, J., Agard, P., Whitechurch, H., Benoit, M., Prouteau, G. & Jolivet, L. 2008. Arc magmatism and subduction history beneath the Zagros Mountains, Iran: A new report of adakites and geodynamic consequences. *Lithos*, **106**, 380–398, https://doi.org/10.1016/j.lithos.2008.09.008

Omrani, H., Moazzen, M., Oberhansli, R., Altenberger, U. & Lange, M. 2013. The Sabzevar blueschists of the North–Central Iranian micro-continent as remnants of the Neotethys-related oceanic crust subduction. *International Journal of Earth Sciences*, **102**, 1491–1512, https://doi.org/10.1007/s00531-013-0881-9

Pang, K.N., Chung, S.L. *et al.* 2013. Eocene–Oligocene post-collisional magmatism in the Lut–Sistan region, eastern Iran: Magma genesis and tectonic implications. *Lithos*, **180–181**, 234–251, https://doi.org/10.1016/j.lithos.2013.05.009

Pearce, J.A. 1982. Trace element characteristics of lavas from destructive plate boundaries. *In*: Thorpe, R.S. (ed) *Andesites: Orogenic Andesites and Related Rocks*. Wiley, Chichester, 525–548.

Pearce, J.A. 2008. Geochemical fingerprinting of oceanic basalts with applications to ophiolite classification and the search for Archean oceanic crust. *Lithos*, **100**, 14–48, https://doi.org/10.1016/j.lithos.2007.06.016

Pearce, J.A. & Robinson, P.T. 2010. The Troodos ophiolitic complex probably formed in a subduction initiation, slab edge setting. *Gondwana Research*, **18**, 60–81, https://doi.org/10.1016/j.gr.2009.12.003

Pearce, J.A., Harris, N.B.W. & Tindle, A.G. 1984. Trace element discrimination diagrams for the tectonic interpretation of granitic rocks. *Journal of Petrology*, **25**, 956–983, https://doi.org/10.1093/petrology/25.4.956

Petrone, C.M. & Ferrari, L. 2008. Quaternary adakite–Nb-enriched basalt association in the western Trans-Mexican volcanic belt: is there any slab melt evidence? *Contributions to Mineralogy and Petrology*, **156**, 73–86, https://doi.org/10.1007/s00410-007-0274-9

Pilger, A. 1971. Die zeitlich-tektonische Entwicklung der iranischen Gebirge, *Clausther Geologische Adhandlunger, Clausthal*, **8**.

Plank, T. 2005. Constraints from thorium/lanthanum on sediment recycling at subduction zones and the evolution of the continents. *Journal of Petrology*, **46**, 921–944, https://doi.org/10.1093/petrology/egi005

Richards, J. & Sholeh, A. 2016. The Tethyan tectonic history and Cu–Au metallogeny of Iran. *In*: Richards, J.P. (ed.) *Tectonics and Metallogeny of the Tethyan Orogenic Belt*. Society of Economic Geologists, Special Publication, **19**, 193–212, https://doi.org/10.5382/SP.19.07

Ricou, L.E. 1968. Une coupe à travers les séries à radiolarites des Monts Pichakun (Zagros, Iran). *Bulletin de la Société Géologique de France*, **10**, 478–485.

Rossetti, F., Nasrabady, M., Vignaroli, G., Theye, T., Gerdes, A., Razavi, M. & Moin Vazir, H. 2010. Early Cretaceous migmatitic mafic granulites from the Sabzevar range (NE Iran): implications for the closure of the Mesozoic peri-Tethyan oceans in central Iran. *Terra Nova*, **22**, 26–34, https://doi.org/10.1111/j.1365-3121.2009.00912.x

Sengor, A.M.C., Altiner, D., Cin, A., Ustaomer, T. & Hsu, K.J. 1988. The Tethyside orogenic collage. *In*: Audley-Charles, M.G. & Hallam, A. (eds) *Gondwana and Tethys*. Geological Society, London, Special Publications, **37**, 119–181.

Shabanian, E., Acocella, V., Gioncada, A., Ghasemi, H. & Bellier, O. 2012. Structural control on volcanism in intraplate post collisional settings: Late Cenozoic to Quaternary examples of Iran and Eastern Turkey. *Tectonics*, **31**, 3013–3042, https://doi.org/10.1029/2011TC003042

Shojaat, B., Hassanipak, A.A., Mobasher, K. & Ghazi, A.M. 2003. Petrology, geochemistry and tectonics of the Sabzevar ophiolite, North Central Iran. *Journal of Asian Earth Sciences*, **21**, 1053–1067, https://doi.org/10.1016/S1367-9120(02)00143-8

Spies, O., Lensch, G. & Mihm, A. 1983. Geochemistry of the post-ophiolitic Tertiary volcanics between Sabzevar and Quchan (NW Iran). Geodynamic project (Geotraverse) in Iran. Geological Survey of Iran, Report, **51**, 264–289.

Stacey, J.S. & Kramers, J.D. 1975. Approximation of terrestrial lead isotope evolution by a two-stage model. *Earth and Planetary Science Letters*, **26**, 207–221, https://doi.org/10.1016/0012-821X(75)90088-6

Sun, S. & McDonough, W.F. 1989. Chemical and isotopic systematics of oceanic basalts: Implications for mantle composition and processes. *In*: Saunders, A.D. & Norry, M.J. (eds) *Magmatism in the Ocean Basins*. Geological Society, London, Special Publications, **42**, 313–345, https://doi.org/10.1144/GSL.SP.1989.042.01.19

Wang, Y.J., Zhang, A.M., Fan, W.M. & Zhang, Y.H. 2013. Origin of paleosubduction-modified mantle for Silurian gabbro in the Cathaysia Block: geochronological and geochemical evidence. *Lithos*, **160**, 37–54, https://doi.org/10.1016/j.lithos.2012.11.004

Woodhead, J.D. & Hergt, J.M. 2005. A preliminary appraisal of seven natural zircon reference materials for *in situ* Hf isotope determination. *Geostandards and Geoanalytical Research*, **29**, 183–195, https://doi.org/10.1111/j.1751-908X.2005.tb00891.x

Woodhead, J., Eggins, S. & Gamble, J. 1993. High field strength and transition element systematics in island arc and back-arc basin basalts: evidence for multi-phase melt extraction and a depleted mantle wedge. *Earth and Planetary Science Letters*, **114**, 491–504, https://doi.org/10.1016/0012-821X(93)90078-N

Xu, Y., Jiang, S., Zhu, Z., Yang, S. & Zhou, W. 2014. Petrogenesis of Late Mesozoic granitoids and coeval mafic rocks from the Jiurui district in the Middle–Early Yangtze metallogenic belt of Eastern China: Geochemical and Sr–Nd-isotopic evidence. *Lithos*, **190–191**, 467–484, https://doi.org/10.1016/j.lithos.2013.12.022

Yousefi, F., Sadeghian, M., Samyari, S. & Ghasemi, H. 2017. Geochemistry and tectonic setting of high silica adakitic domes of Ahmad Abad Khartouran (Southeast Shahrood). *Geosciences*, **25**, 239–252.

Yuan, H.L., Gao, S. *et al.* 2008. Simultaneous determinations of U–Pb age, Hf isotopes and trace element compositions of zircon by excimer laser-ablation quadrupole and multiple-collector ICP-MS. *Chemical Geology*, **247**, 100–118, https://doi.org/10.1016/j.chemgeo.2007.10.003